

SDEV2501

Lesson 16 – Lying with Visualization

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

Today’s objective

- identify and explain common ways visualizations can mislead audiences.

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Agenda

- Review examples of misleading visualizations
- Identify what impression it creates & how they can be improved

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REVIEW

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Data Viz

- What are examples of data viz for continuous data?
  - For discrete data?

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LYING WITH VISUALIZATION

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Habits we want to build

- Inspect the **encoding**: What do the axes, scales, colours, sizes, and visual elements actually represent?
- Inspect the **evidence**: What data, context, comparison, or uncertainty is missing?
- Inspect the **claim**: Does the conclusion follow from the evidence, or is the visualization encouraging us to believe something stronger?

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Note that “lying” isn’t always intentional.

A misleading visualization isn't **necessarily** an intentional lie.

It can result from

- poor design,
- an analytical mistake,
- selective reporting, or
- deliberate manipulation.

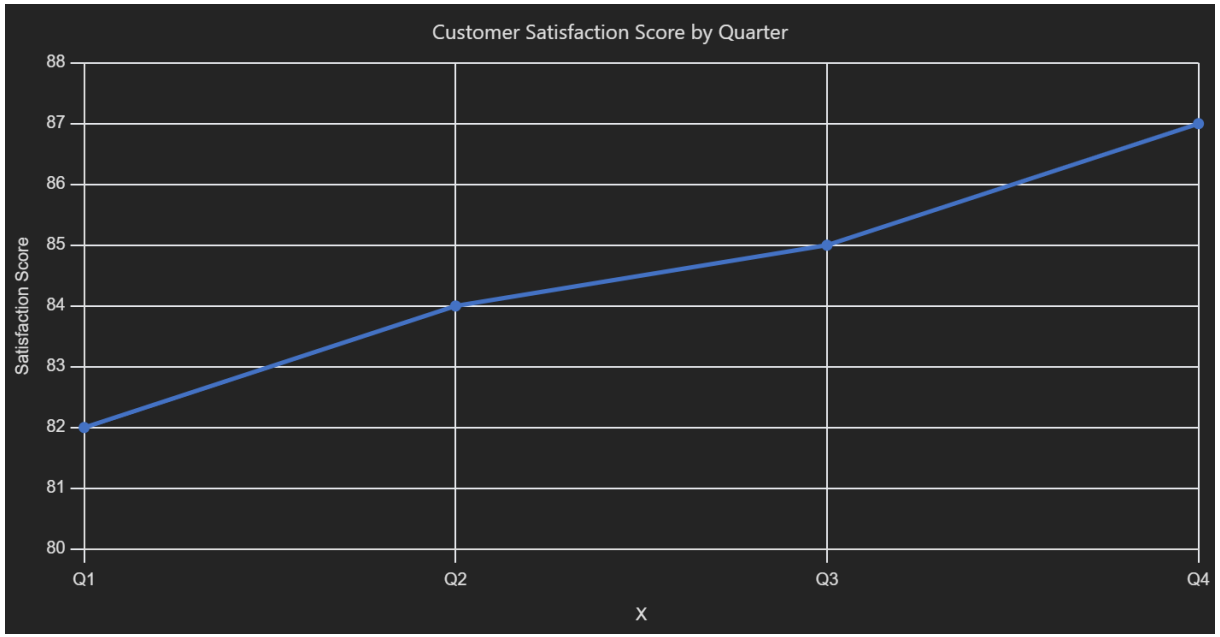
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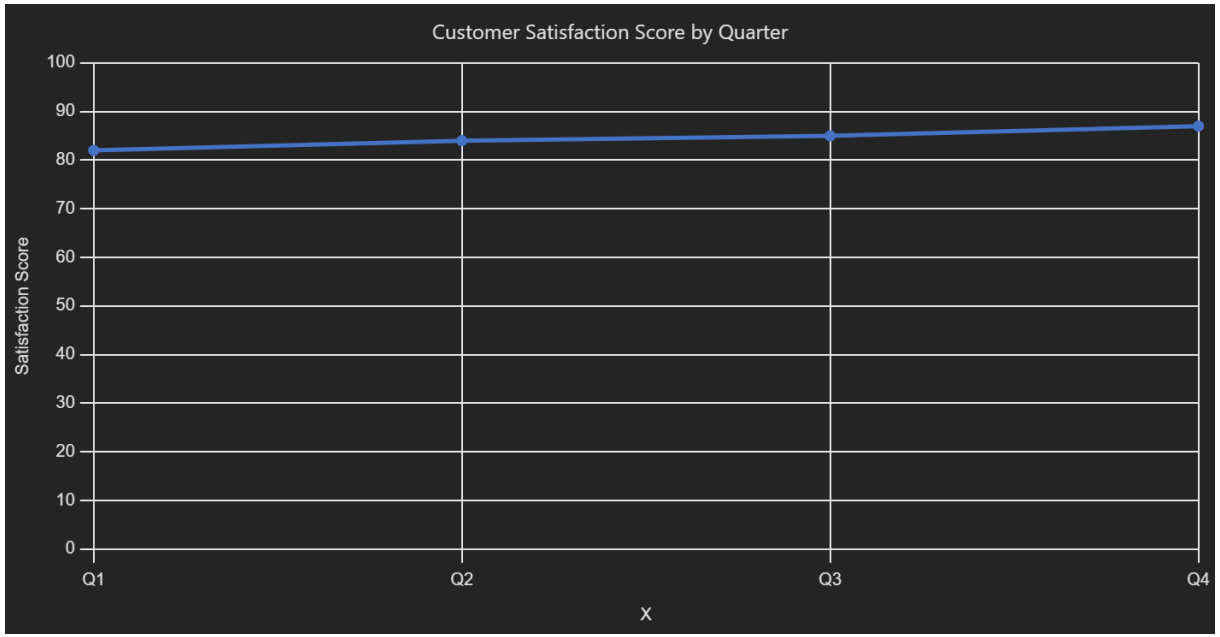
Introduction

- A visualization can be technically accurate but still create a **misleading** impression.
- Misleading design can affect what viewers notice, how large they perceive a difference to be, and what **conclusions** they draw.
- The problem may be in the **chart**, the **data** selection, the **comparison**, the **annotation**, or the **conclusion**.
- Always distinguish between what a chart **shows**, what it **suggests**, and what the **evidence** actually supports.
- Before criticizing a chart, identify the **question** it is **supposed** to answer.

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What makes this chart dramatic?





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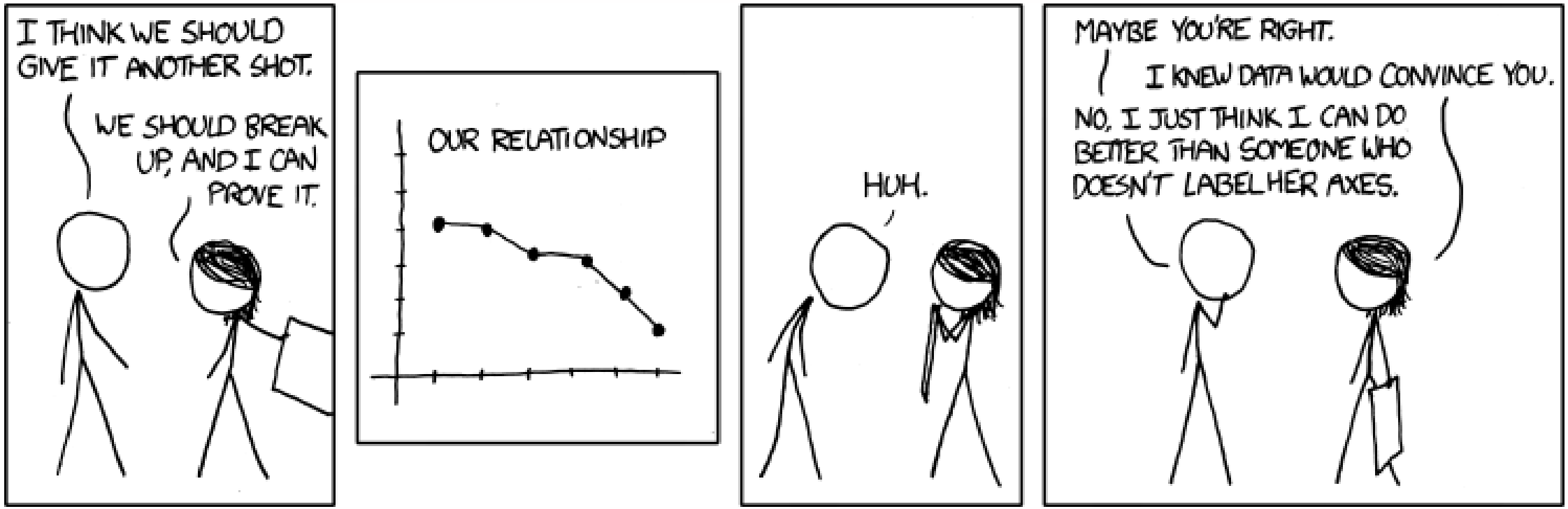
WHAT NOT TO DO

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What not to do:

FORGET TO ANNOTATE.

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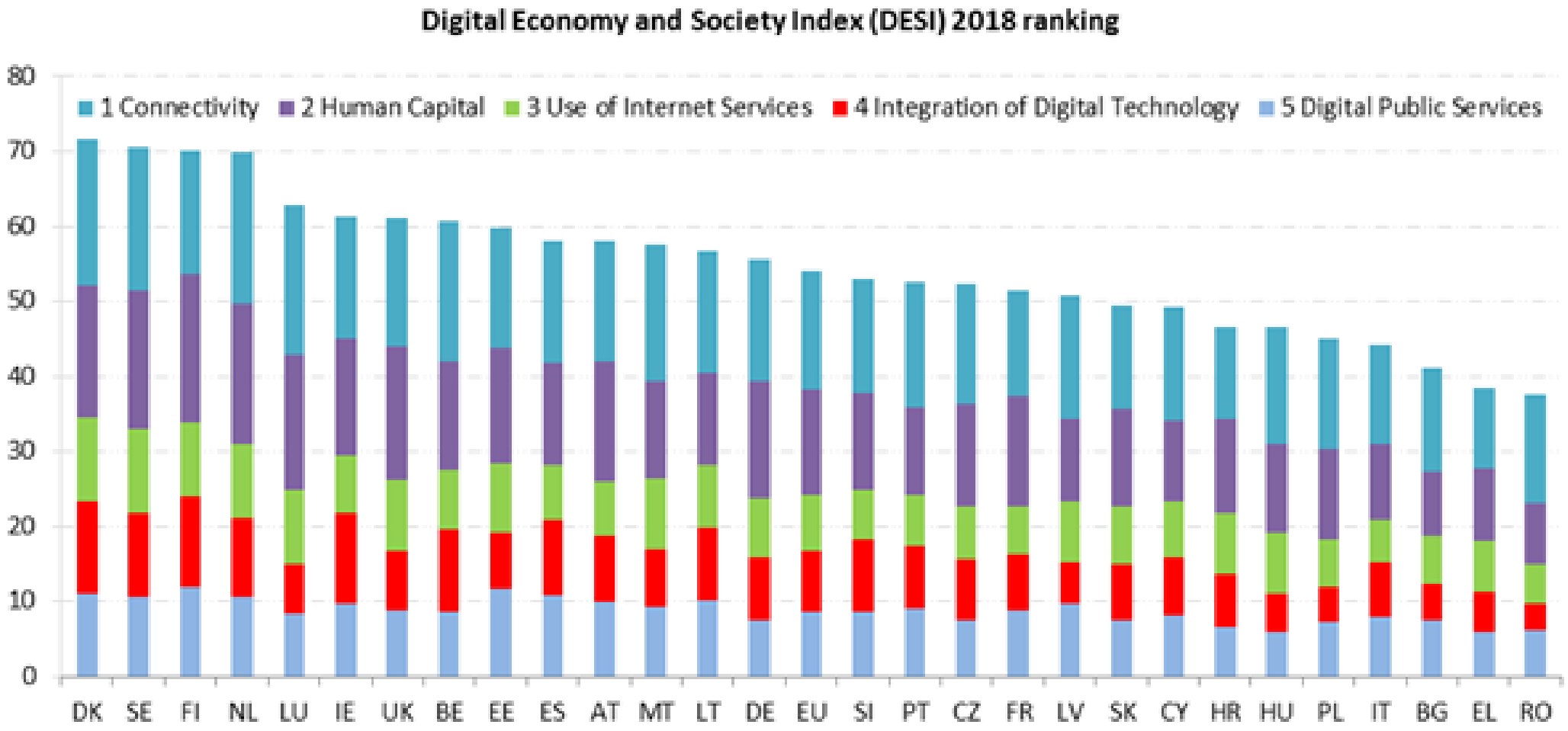

I THINK WE SHOULD GIVE IT ANOTHER SHOT.
WE SHOULD BREAK UP, AND I CAN PROVE IT.
OUR RELATIONSHIP
HUH.
MAYBE YOU'RE RIGHT.
I KNEW DATA WOULD CONVINCE YOU.
NO, I JUST THINK I CAN DO BETTER THAN SOMEONE WHO DOESN'T LABEL HER AXES.

What not to do:

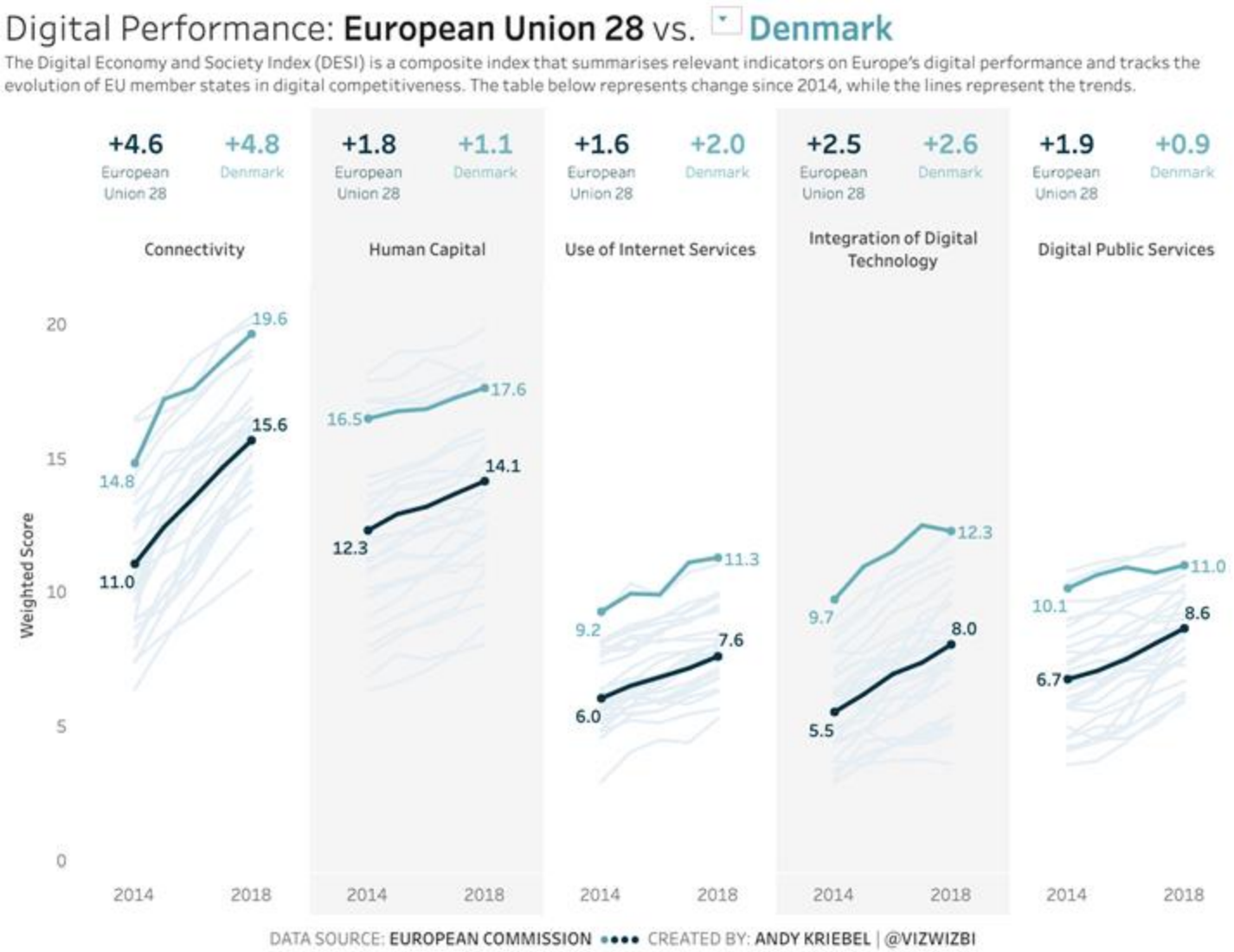
GIVE YOUR VIEWER A HEADACHE.

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This is bad.


Digital Economy and Society Index (DESI) 2018 ranking
1 Connectivity
2 Human Capital
3 Use of Internet Services
4 Integration of Digital Technology
5 Digital Public Services
80
70
60
50
40
30
20
10
0
DK SE FI NL LU IE UK BE EE ES AT MT LT DE EU SI PT CZ FR LV SK CY HR HU PL IT BG EL RO

This is better.

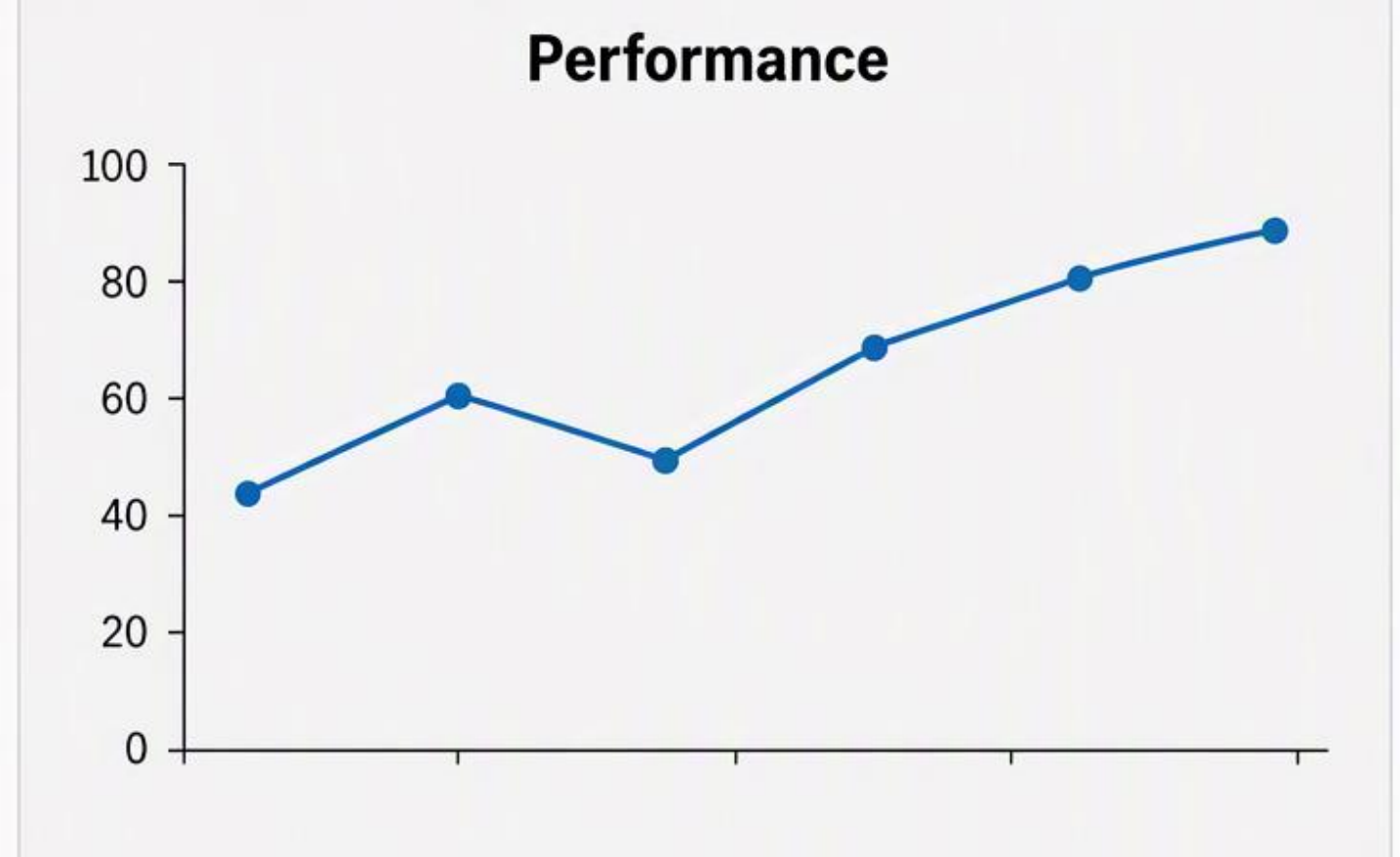


Missing annotations & unnecessary complexity

- Every chart should communicate what is being measured, the relevant units, and the time period or population covered.
- Legends, labels, and annotations should make the chart easier to interpret, not force the audience to guess.
- Too many categories, colours, labels, or stacked segments can obscure the pattern that matters.
- Decorative elements should not compete with the data.
- A chart should make the important comparison easy to identify without requiring the audience to reconstruct the dataset.

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Practice: the mystery dashboard



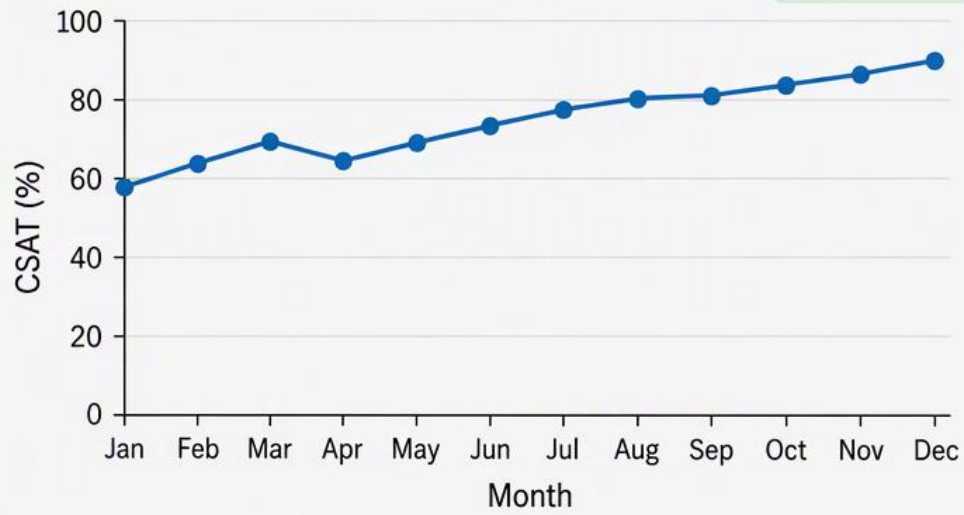
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Improved dashboard

Customer Satisfaction Score (CSAT)

Jan 2024 – Dec 2024

↑ Higher scores are better



What's improved?

- **Specific title:** shows what is being measured
- **Clear axis labels:** includes metric name and units (percentage)
- **Time period:** Jan 2024 – Dec 2024
- **Context:** higher values indicate better performance
- **Note:** defines the metric

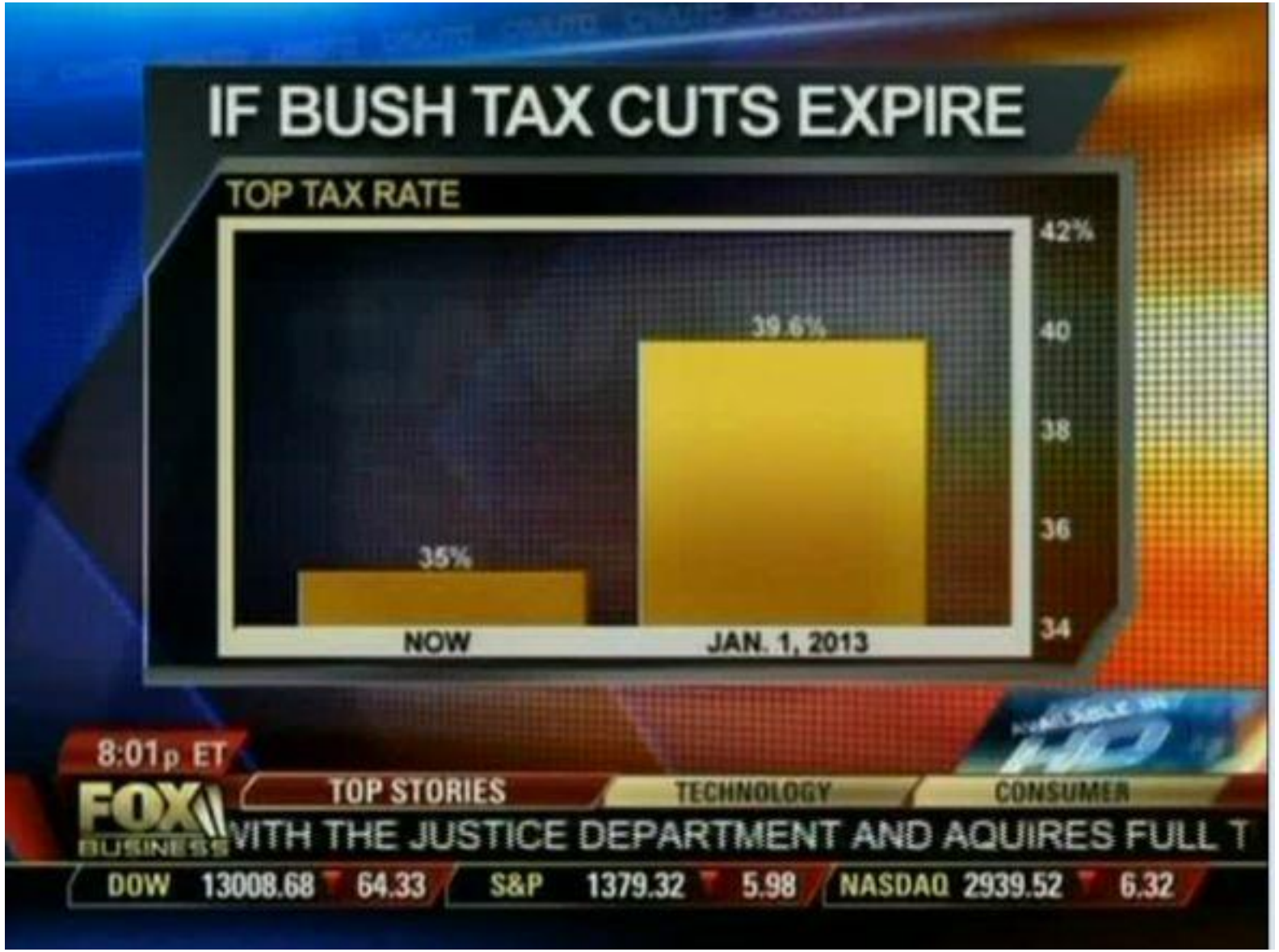
Note: Customer Satisfaction Score (CSAT) is the percentage of customers who rated their experience 4 or 5 out of 5 in our post-purchase survey.

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What not to do:

DISTORT.

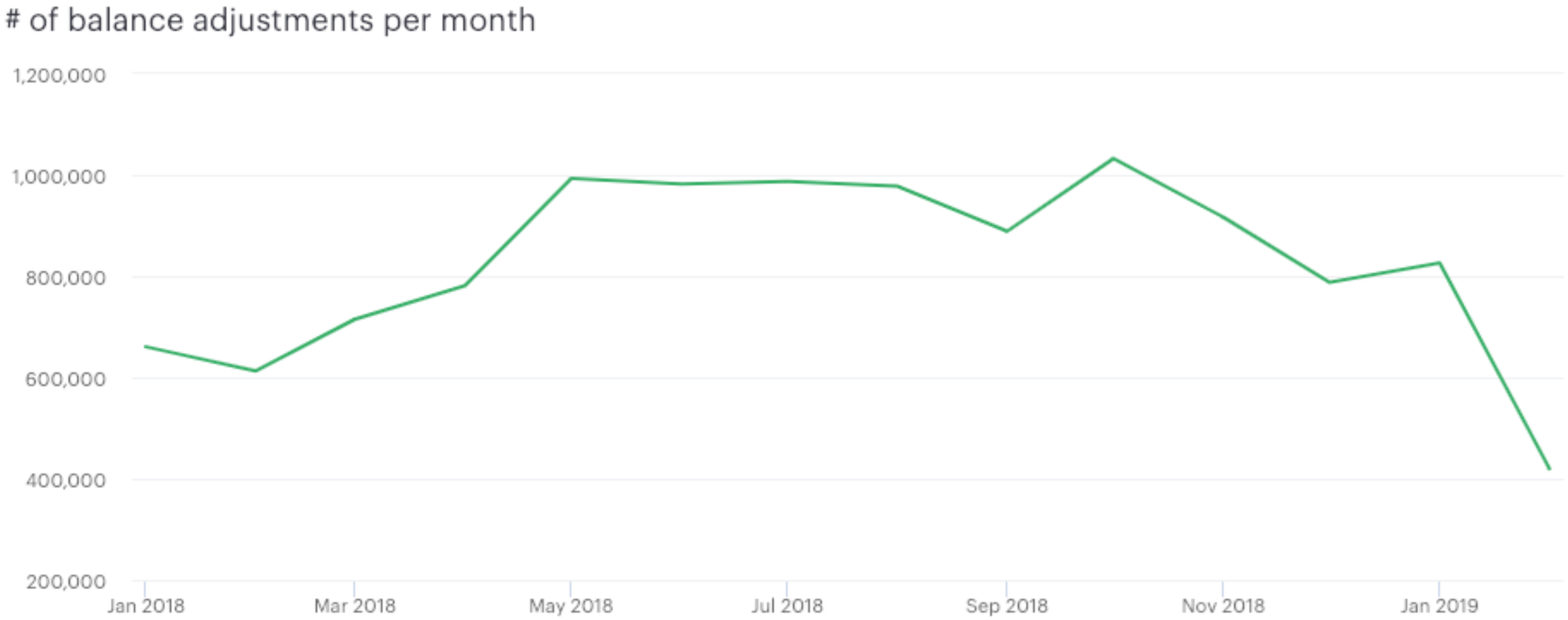
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IF BUSH TAX CUTS EXPIRE
TOP TAX RATE
42%
40
38
36
34
39.6%
35%
NOW
JAN. 1, 2013
8:01p ET
FOX BUSINESS
TOP STORIES
TECHNOLOGY
CONSUMER
VITH THE JUSTICE DEPARTMENT AND AQUIRES FULL T
DOW 13008.68 64.33
S&P 1379.32 5.98
NASDAQ 2939.52 6.32

Tip:

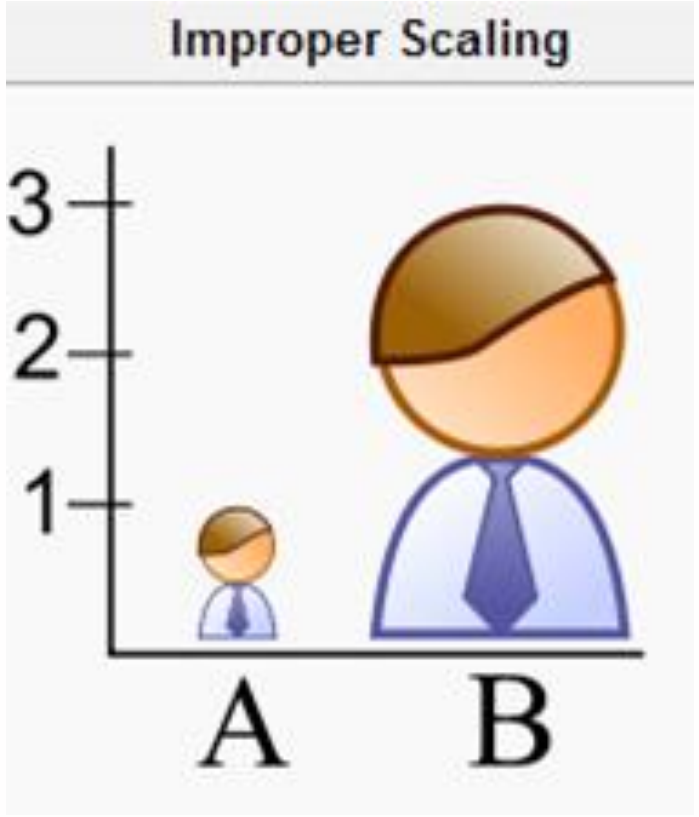
Area/bar charts should ALWAYS start at zero.

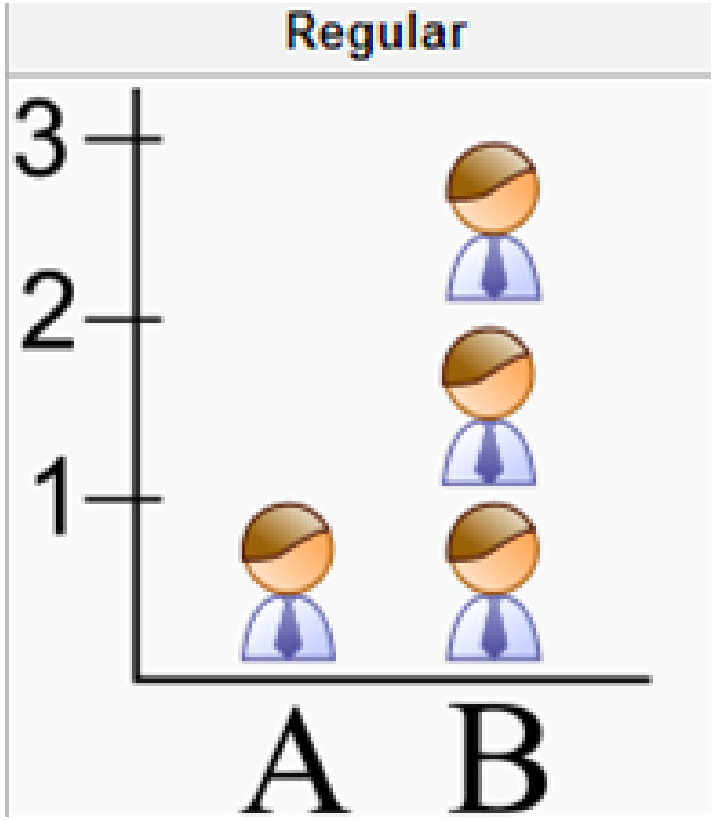
Pulling “monthly” data on Feb 15:

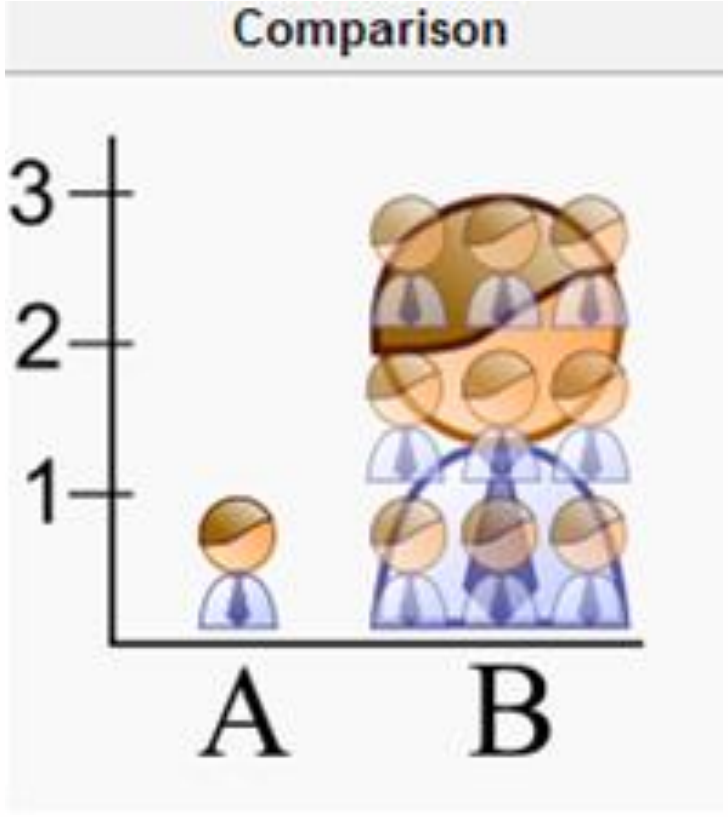


Tip:

Your data should always be in equivalent bins.


Improper Scaling
3
2
1
A
B


Regular
3
2
1
A
B


Comparison
3
2
1
A
B

Tip:

Pick your dimensions and stick to them.

Distorted scale & unfair comparisons

- A truncated axis can make a small difference appear dramatic, particularly in a bar chart.
- A bar chart comparing magnitudes should generally start at zero because bar length represents the value.
- A line chart may use a non-zero y-axis when the goal is to examine variation within a narrow range, but the scale must be clear and the magnitude of the changes must not be overstated.
- Comparisons must use equivalent time periods, definitions, units, and populations.
- Comparing a complete month with an incomplete month can create an apparent decline that disappears once the periods are made comparable.
- Pictograms can exaggerate differences when both height and width change, because the area grows in two dimensions.
- Always inspect the scale before interpreting the apparent size of a difference.

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Example: The dramatic sales increase

- Suppose monthly sales rise from $95,000 to $100,000.
- The increase is $5,000, or approximately 5.3%.
- A chart with a y-axis running from $90,000 to $101,000 may make the increase look enormous.
- A bar chart beginning at zero communicates the relative magnitudes more faithfully.
- A line chart with a tighter scale can still be useful for showing small fluctuations, provided the axis is visible and the interpretation is appropriately qualified.
- Important nuance: A zero baseline is not a universal rule for every visualization. The correct scale depends on the visual encoding and analytical purpose. The key is to avoid concealing the scale or encouraging an interpretation the data does not support.

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Comparison Checklist

Before comparing two values, ask:

- Are the periods the same length?
- Are the units and definitions consistent?
- Are the groups comparable in size?
- Is the denominator the same?
- Does the axis start at zero where required by the encoding?
- Does the visual difference reflect the numerical difference?

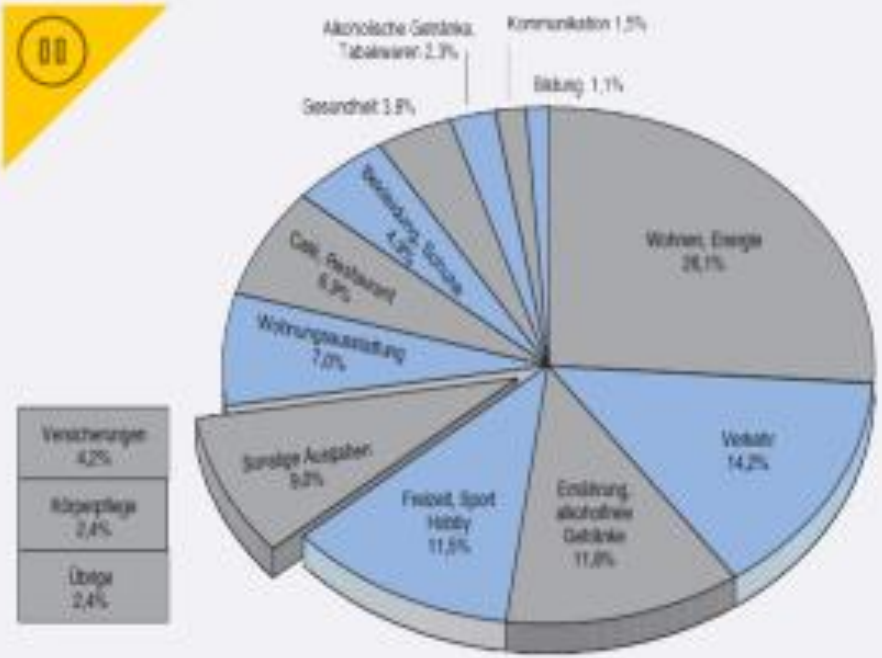
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What not to do:

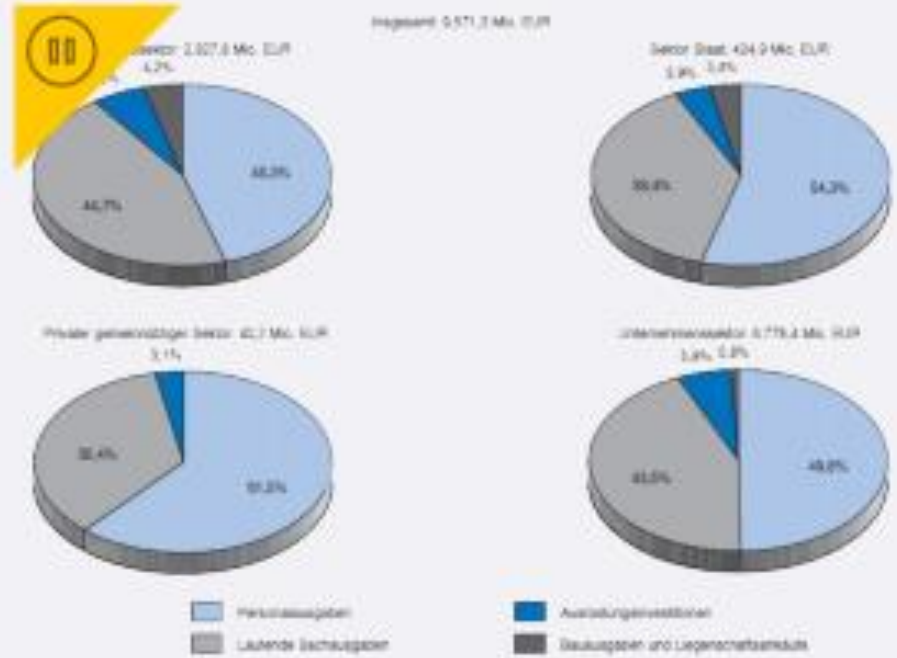
USE 3D.

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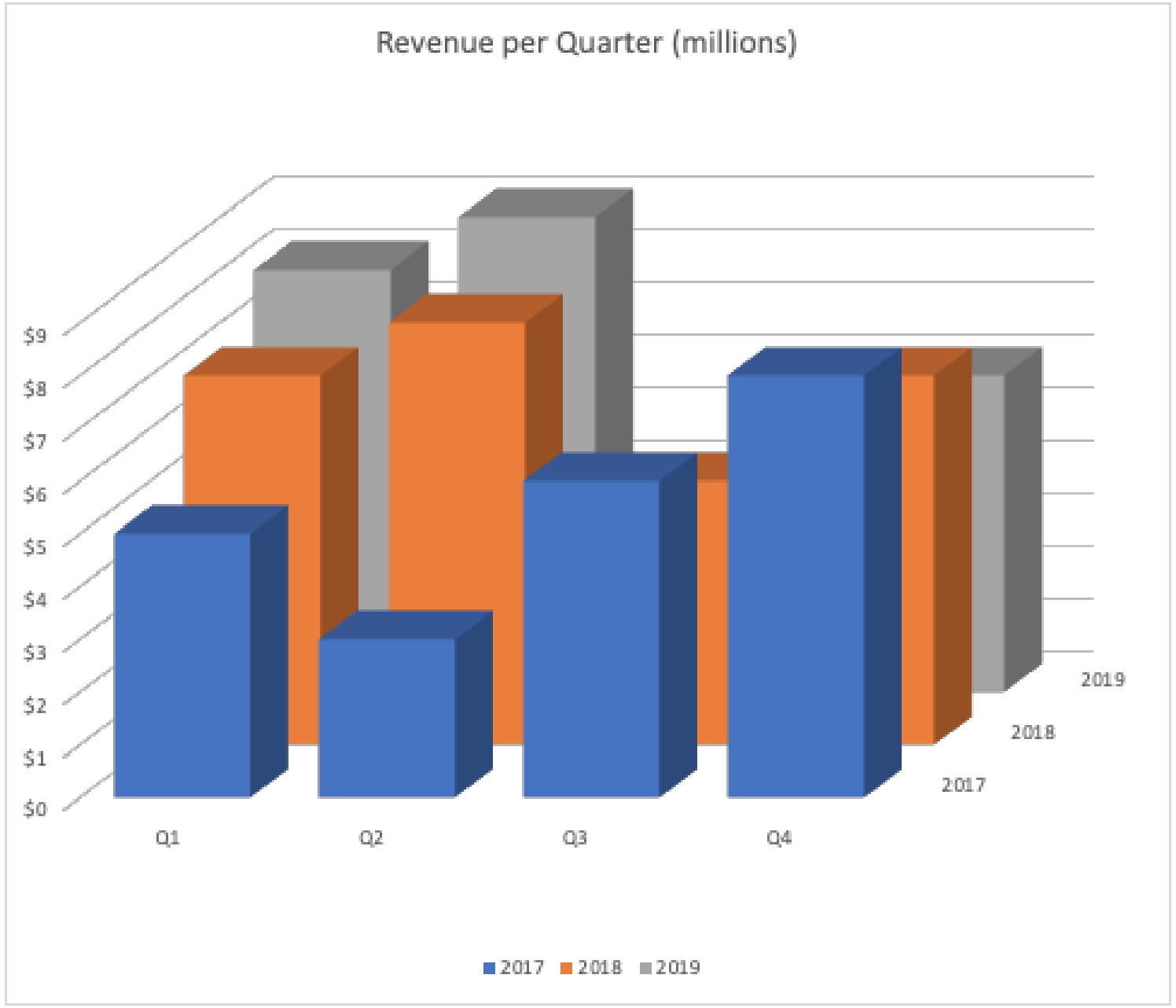
things that belong to the past:
everything 3D



3D pie charts



Comparing 3D pie charts


Revenue per Quarter (millions)
$9
$8
$7
$6
$5
$4
$3
$2
$1
$0
Q1
Q2
Q3
Q4
2017
2018
2019
2017
2018
2019

3D Charts

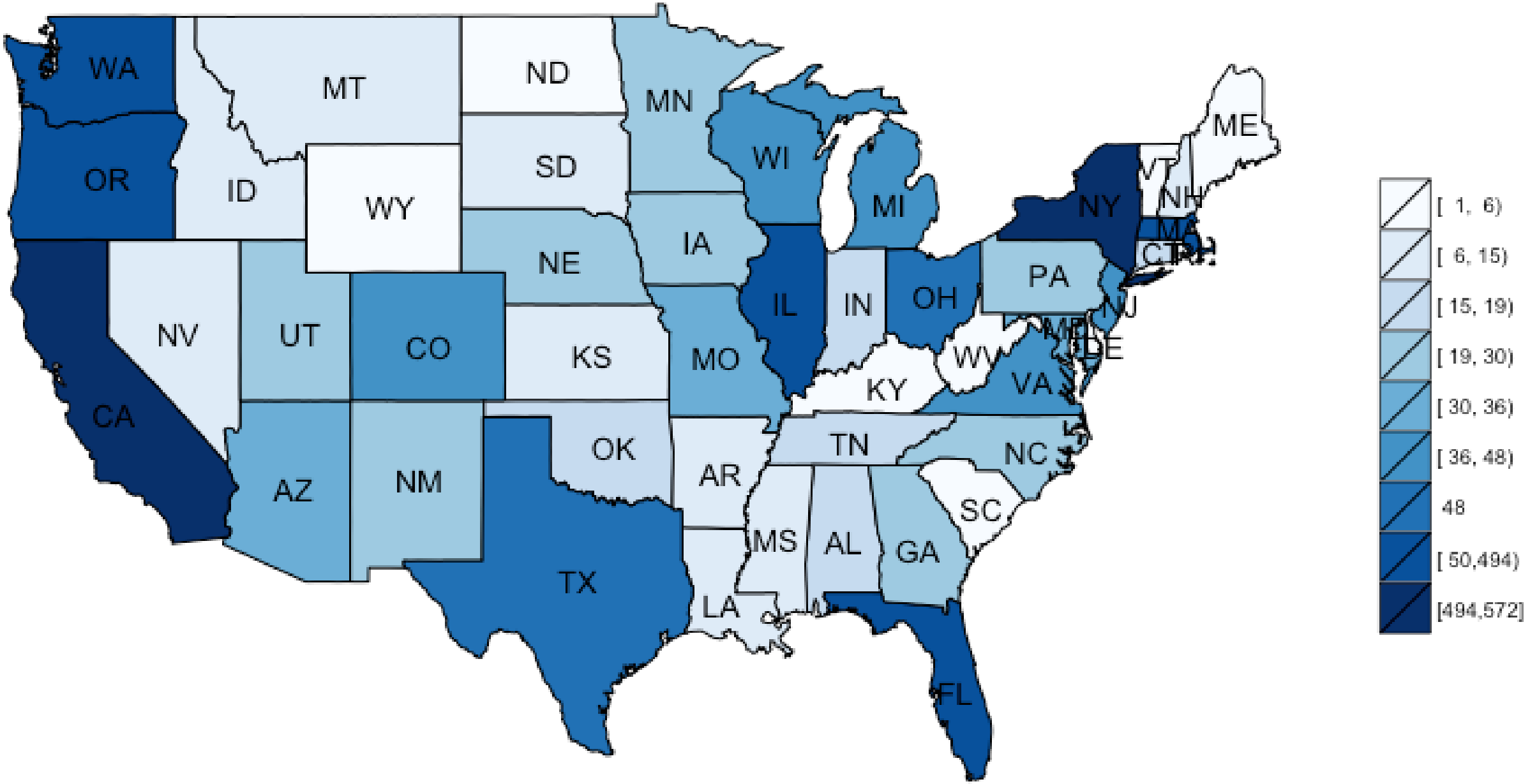
- Perspective changes the apparent size and position of chart elements.
- Front-facing elements may appear larger than equivalent elements at the back.
- Depth makes it harder to compare values accurately.
- Three-dimensional effects rarely add useful information to a simple two-dimensional dataset.
- Use a conventional bar chart, dot plot, or two-dimensional chart instead.
- Quick question: If the purpose of a chart is to compare values accurately, why introduce a visual dimension that does not represent another variable?

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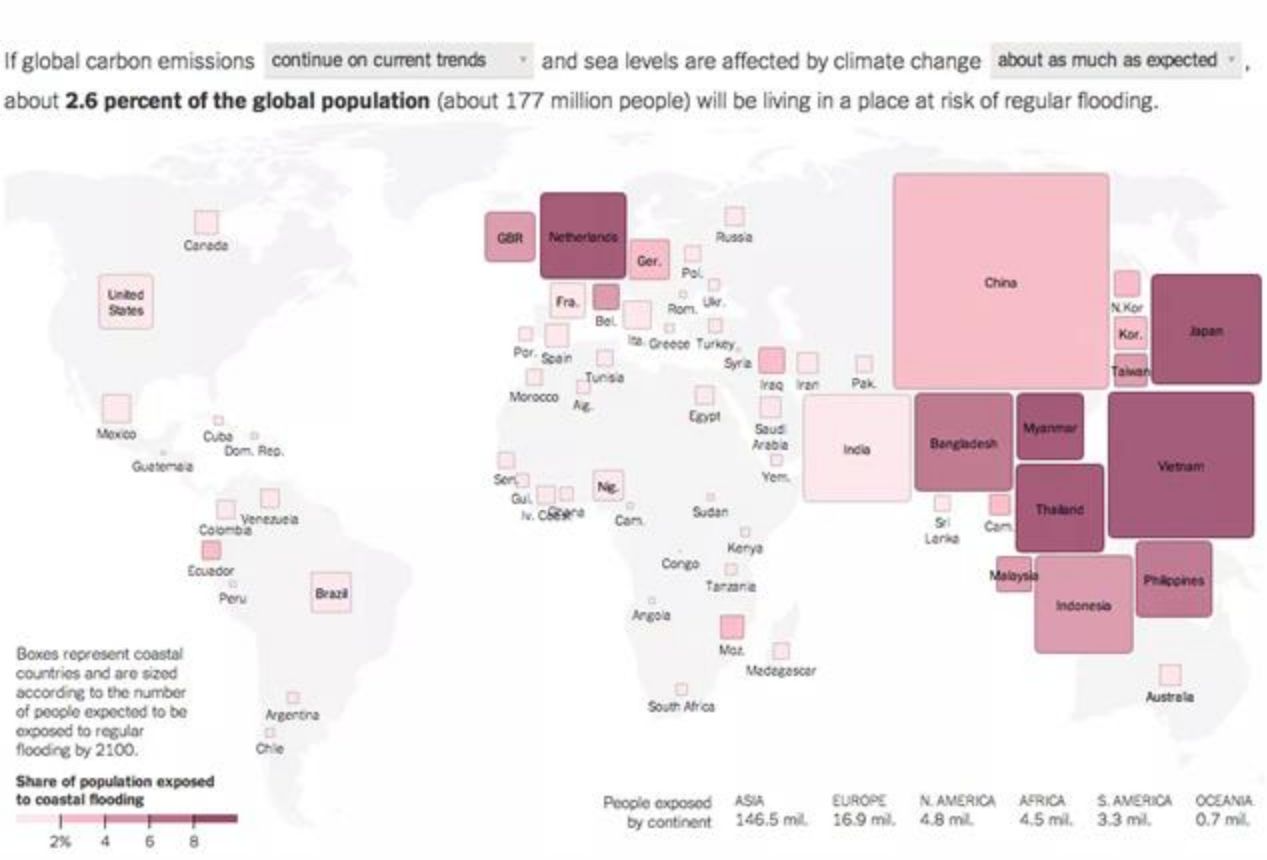
What not to do:

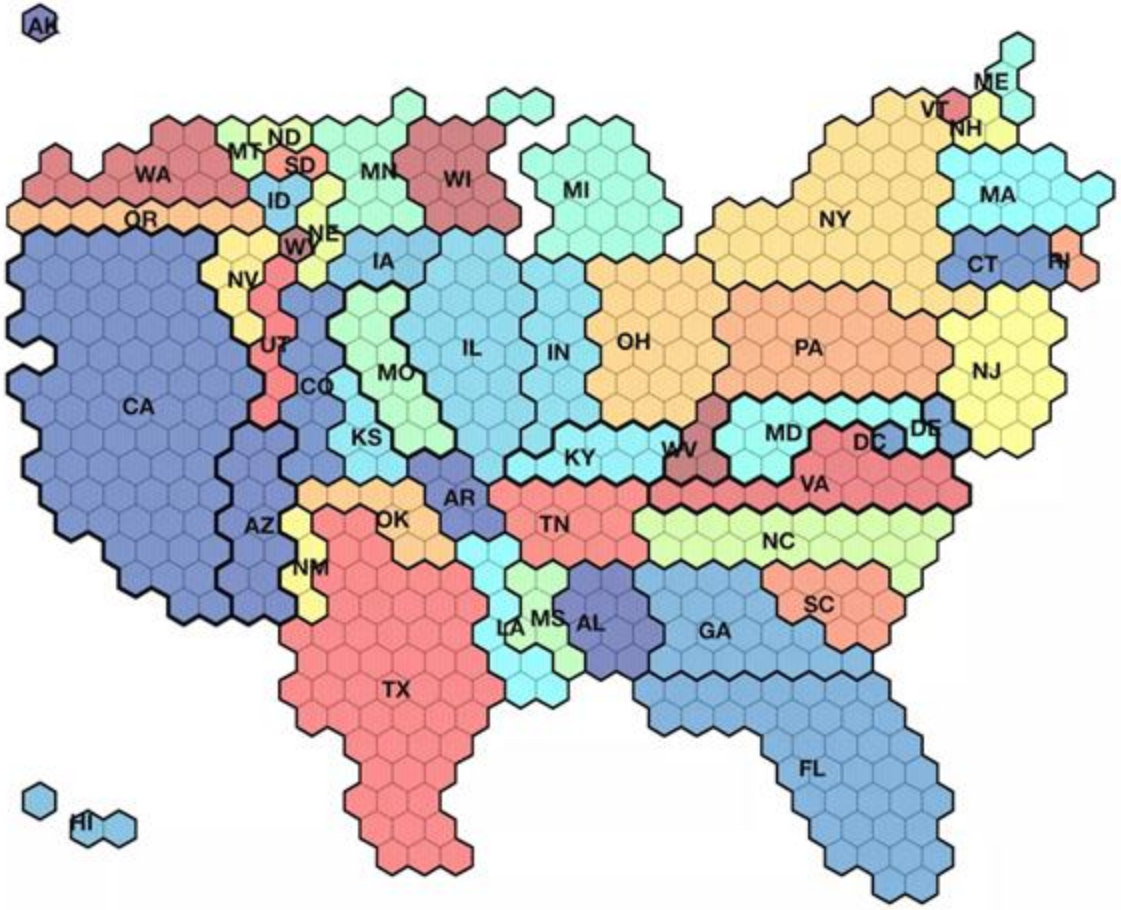
USE CHOROPLETHS FOR ABSOLUTE VALUES.

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WA
OR
CA
ID
NV
MT
WY
UT
AZ
CO
NM
ND
SD
NE
KS
OK
TX
MN
IA
MO
AR
LA
WI
IL
MS
MI
IN
KY
TN
AL
OH
GA
WV
VA
NC
SC
FL
PA
NY
VT
NH
ME
MA
CT
NJ
MD
DE
[1, 6)
[6, 15)
[15, 19)
[19, 30)
[30, 36)
[36, 48)
48
[50,494)
[494,572]

Instead:


If global carbon emissions continue on current trends and sea levels are affected by climate change about as much as expected,
about 2.6 percent of the global population (about 177 million people) will be living in a place at risk of regular flooding.
Canada
United States
Mexico
Cuba
Dom. Rep.
Guatemala
Colombia
Venezuela
Ecuador
Peru
Brazil
Argentina
Chile
GBR
Netherlands
Ger.
Russia
Pol.
Ukr.
Rom.
Fra.
Bel.
Ita.
Greece
Turkey
Por.
Spain
Tunisia
Morocco
Alg.
Egypt
Syria
Iraq
Iran
Pak.
Saudi Arabia
Yem.
Sen.
Gui.
Ivory Coast
Ghana
Nig.
Cam.
Sudan
Kenya
Congo
Tanzania
Angola
Moz.
Madagascar
South Africa
China
India
N.Kor.
Kor.
Taiwan
Japan
Bangladesh
Myanmar
Thailand
Sri Lanka
Cam.
Vietnam
Malaysia
Indonesia
Philippines
Australia
Boxes represent coastal countries and are sized according to the number of people expected to be exposed to regular flooding by 2100.
Share of population exposed to coastal flooding
2%
4
6
8
People exposed by continent
ASIA 146.5 mil.
EUROPE 16.9 mil.
N. AMERICA 4.8 mil.
AFRICA 4.5 mil.
S. AMERICA 3.3 mil.
OCEANIA 0.7 mil.


AK
HI
WA
OR
CA
ID
NV
MT
WY
UT
AZ
CO
NM
ND
SD
NE
KS
OK
TX
MN
IA
MO
AR
LA
WI
IL
MS
MI
IN
KY
TN
AL
OH
WV
GA
FL
SC
NC
VA
PA
NY
MD
DC
DE
NJ
CT
RI
MA
VT
NH
ME

Misleading maps

- A choropleth map uses colour or shading to represent a value across geographic areas.
- Choropleth maps are generally best suited to rates, percentages, densities, or other standardized measures, rather than raw totals.
- Large geographic areas can visually dominate a map even when they contain relatively few people or events.
- When comparing populations of different sizes, consider whether a per-capita rate or another denominator is more meaningful than an absolute count.
- Map projections distort some combination of area, distance, shape, or direction; the appropriate projection depends on the intended use.
- A cartogram deliberately changes geographic area to represent a variable, such as population. It can be useful, but distorts geographic shape and familiarity.

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Example: Which neighbourhood has the biggest problem?

- Imagine a map showing the number of reported incidents in each neighbourhood.
- A neighbourhood with 500 incidents may have a much larger population than one with 100 incidents.
- Which neighbourhood has the most incidents?
- Which has the highest incident rate per 1,000 residents?
- Which measure is most relevant to the decision being made?

Neither counts nor rates are inherently better. The appropriate measure depends on the question, the population at risk, and the context.

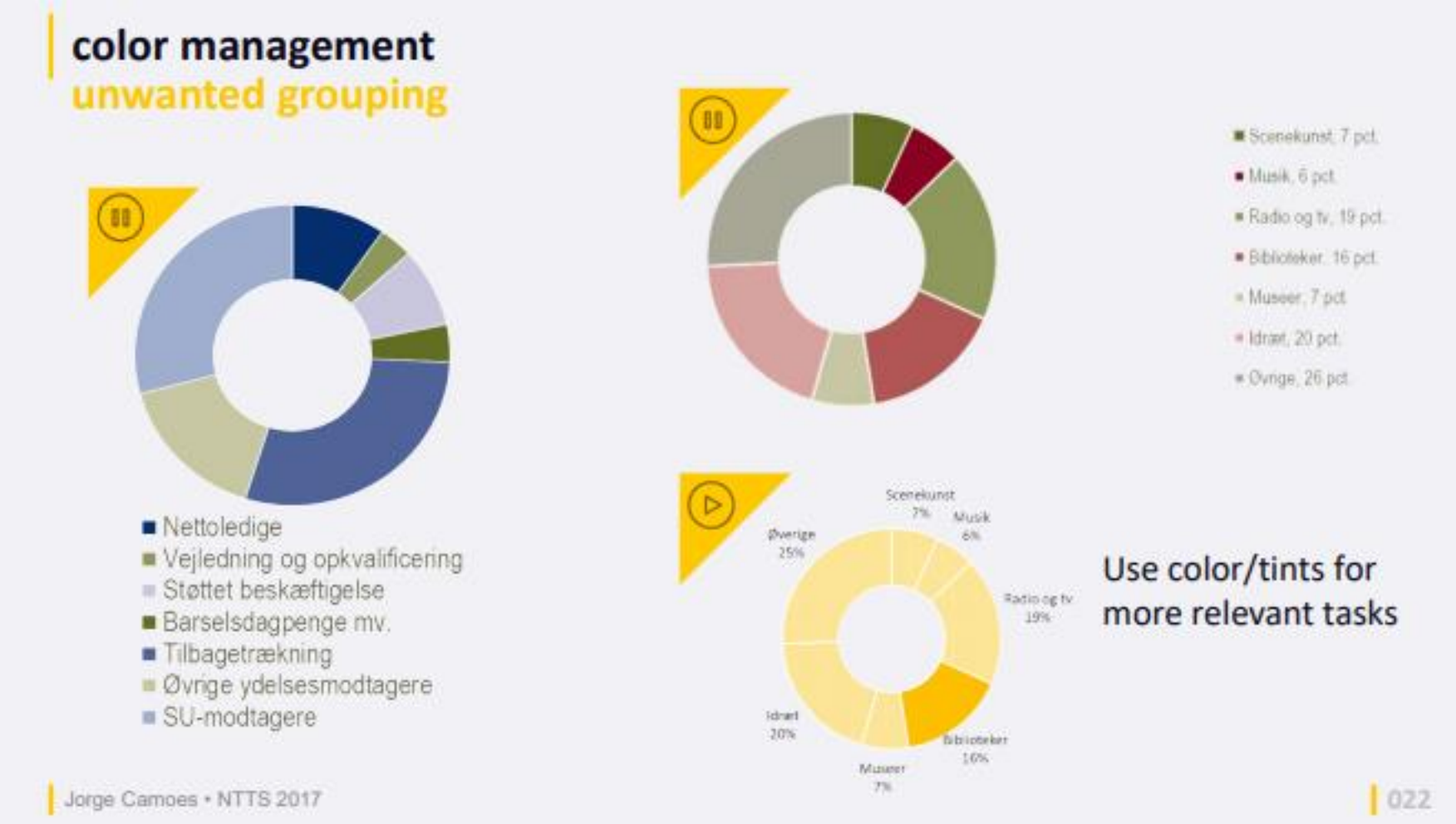
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What not to do:

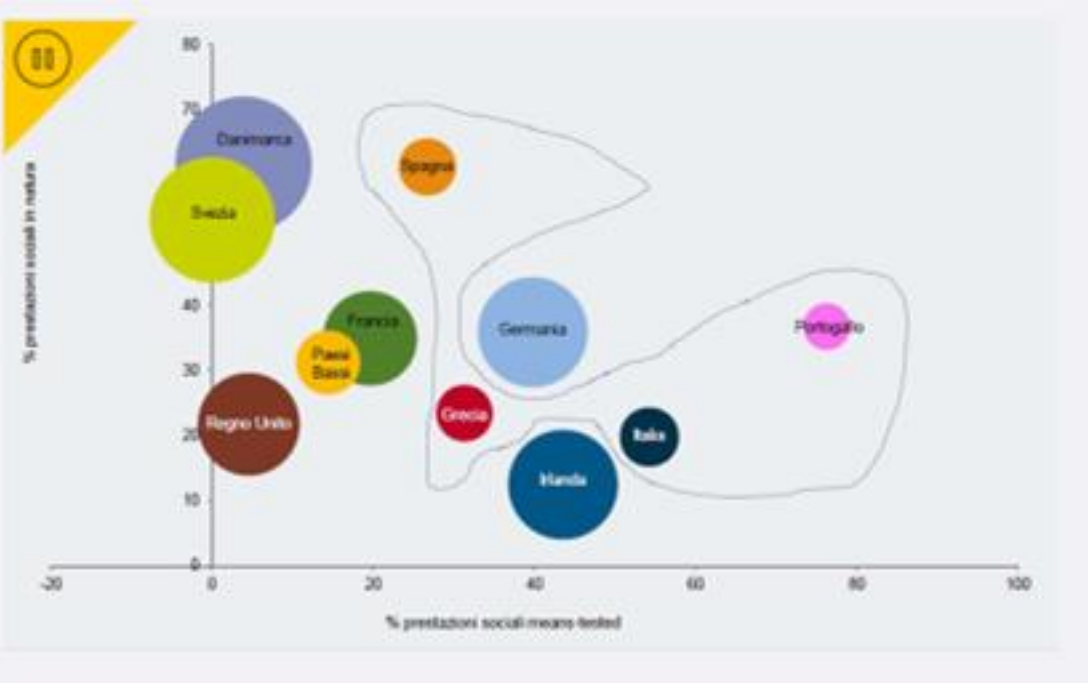
USE COLOUR POORLY.

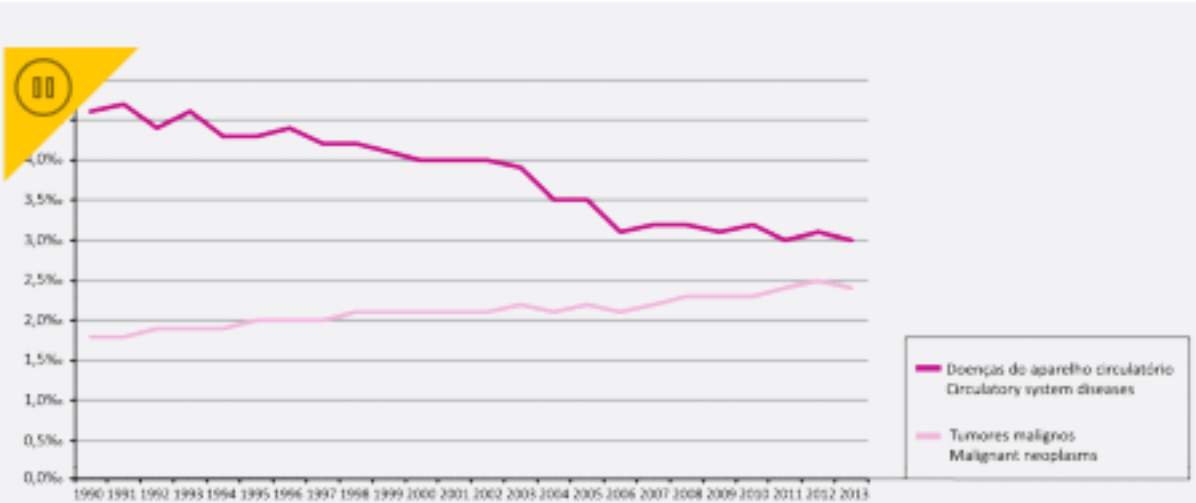
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Do your data belong in groups?



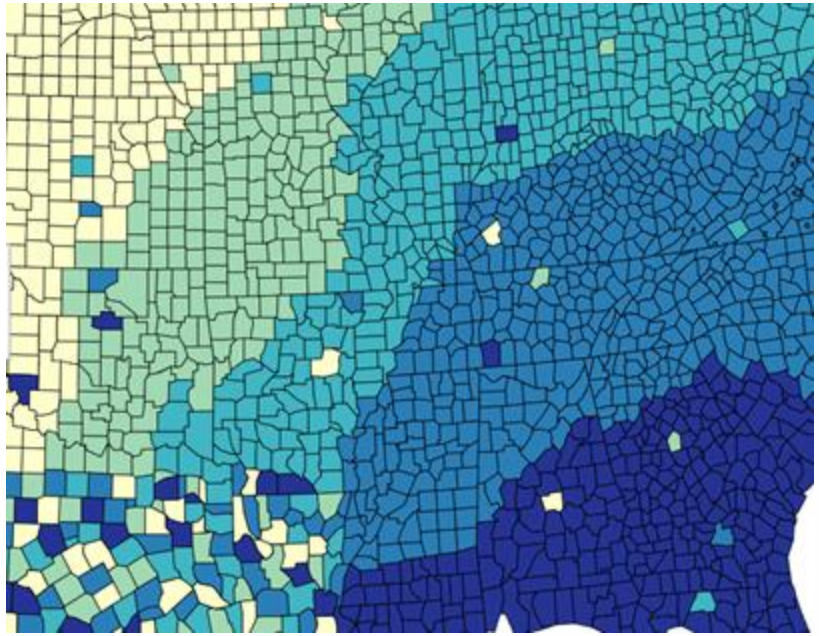
Can you use colour to draw attention?

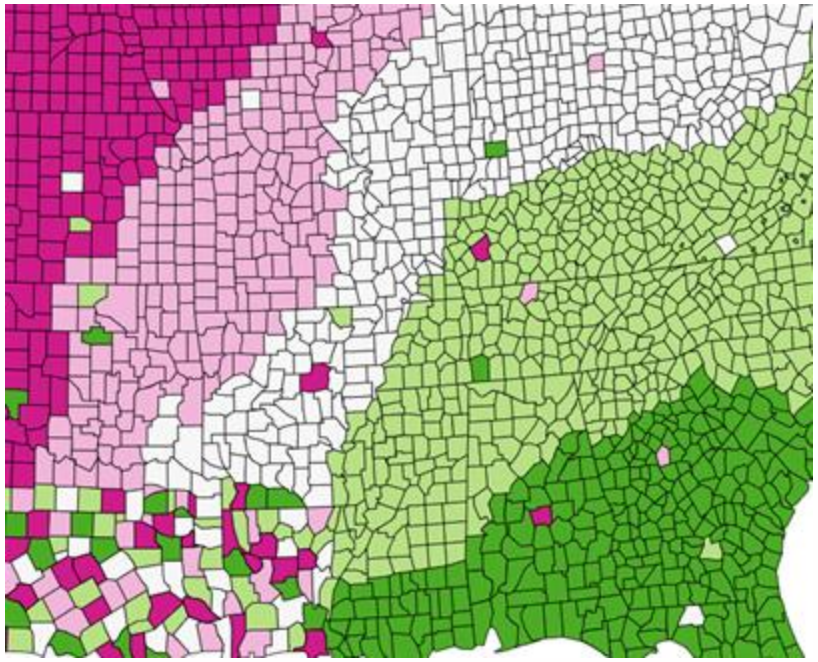


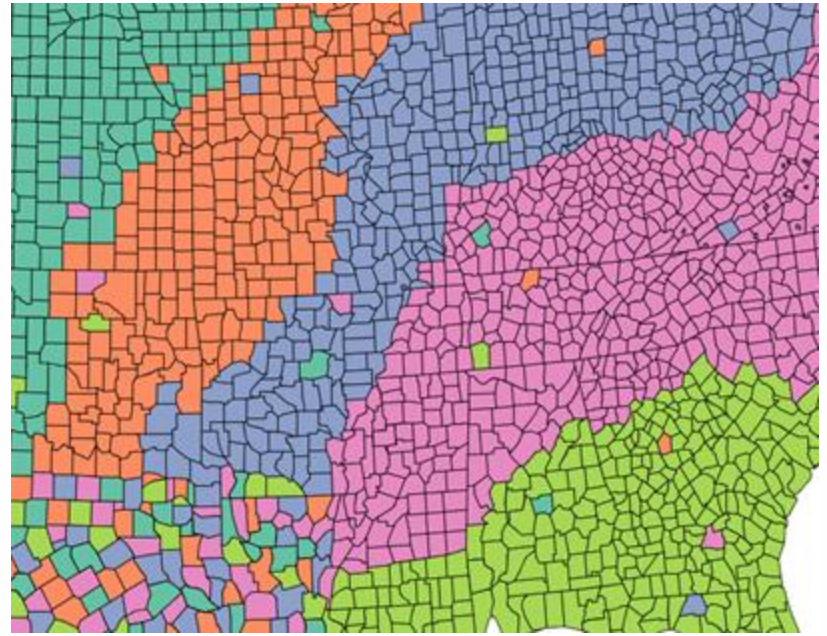


Trend of mortality rate caused by diseases of the circulatory system and malignant neoplasms, 1990 – 2013

Sequential vs diverging vs qualitative palettes







Poor colour choices

- Colour creates a visual hierarchy: brighter, darker, or more saturated elements often attract attention first.
- A colour can unintentionally suggest importance, danger, success, or failure even when those meanings are not justified by the data.
- Using too many unrelated colours can make categories seem more distinct or important than they are.
- Rainbow colour scales can imply boundaries or order that do not correspond to meaningful differences in the data.
- A sequential palette is appropriate for ordered values; a diverging palette is appropriate when deviations from a meaningful midpoint matter; a qualitative palette is suited to distinct categories.
- Colour should reinforce a comparison, not manufacture one.
- Test visualizations for accessibility and avoid using colour as the only indicator of a meaningful difference.

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Example: The highlighted result

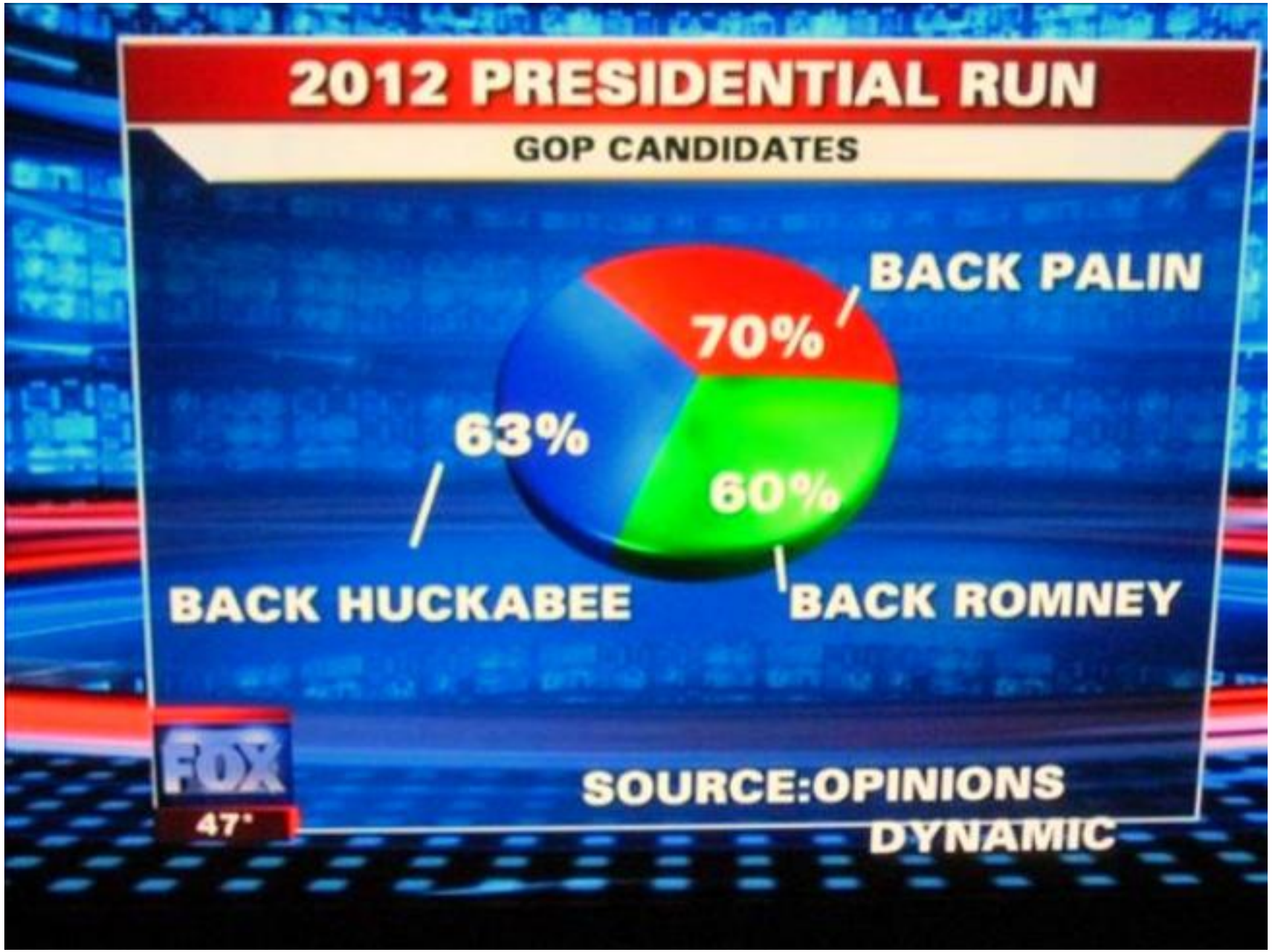
- A dashboard displays five departments in grey and one department in bright red.
- The red department immediately attracts attention.
- Is that department genuinely underperforming?
- Is red used consistently for a defined threshold?
- Would a neutral palette change what you notice first?
- Is the colour directing attention toward the most important result or toward the result the designer wants you to notice?
- Important distinction: Highlighting one value is not automatically deceptive. It becomes problematic when the emphasis creates an unsupported interpretation or distracts from relevant comparisons.

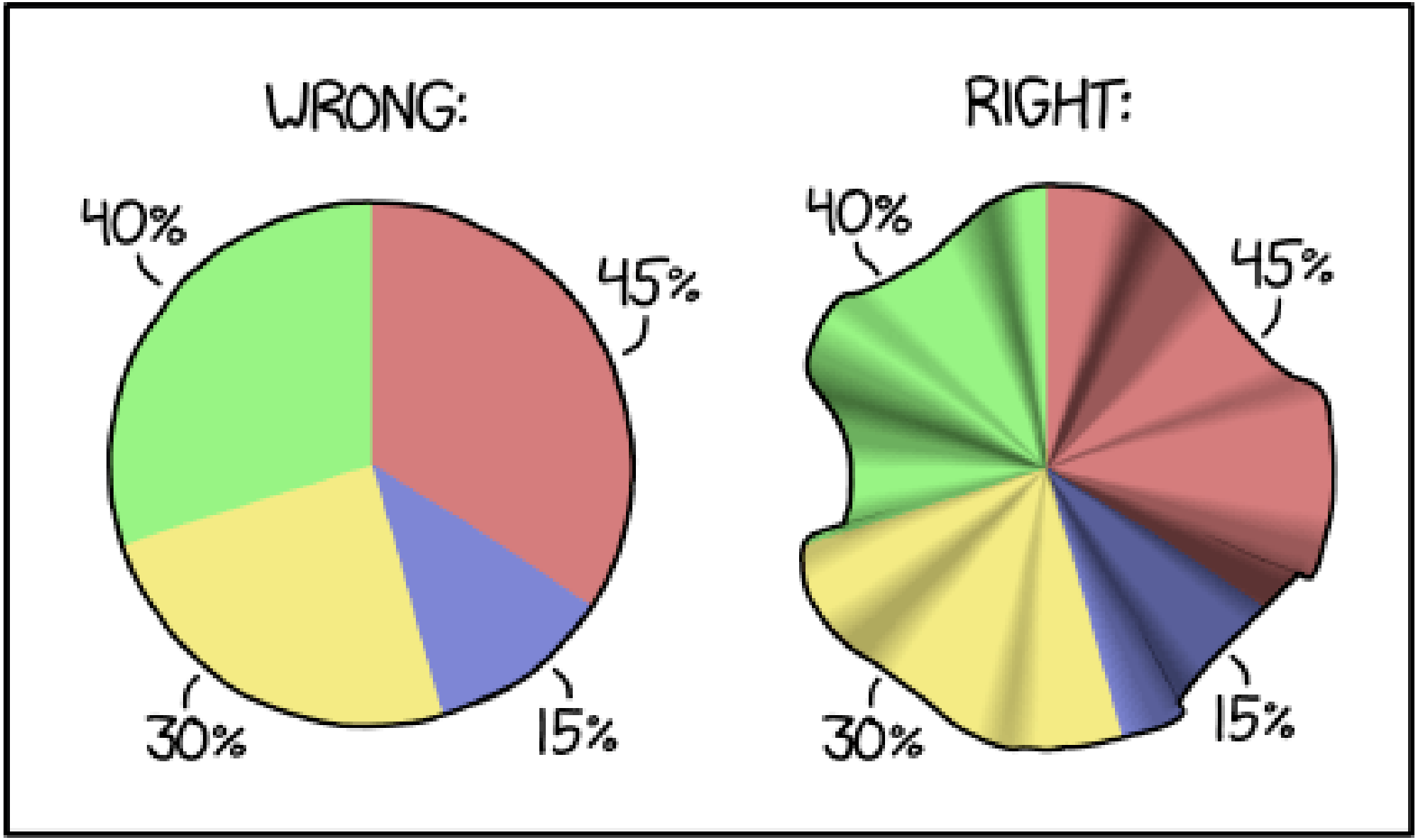
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What not to do:

MAKE NONSENSE PIE CHARTS.

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2012 PRESIDENTIAL RUN
GOP CANDIDATES
BACK PALIN
70%
63%
60%
BACK HUCKABEE
BACK ROMNEY
FOX
47
SOURCE:OPINIONS
DYNAMIC



HOW TO MAKE A PIE CHART IF YOUR PERCENTAGES DON'T ADD UP TO 100

Inappropriate pie charts and percentages

- A pie chart represents parts of a single whole.
- Categories in a conventional pie chart should be mutually exclusive and collectively exhaustive if they are intended to represent the entire whole.
- Percentages for a complete partition should add to approximately 100%, allowing for rounding.
- If categories overlap, the total may legitimately exceed 100% — but a pie chart is not appropriate for representing those overlapping percentages.
- A pie chart with many small segments makes comparisons difficult; a sorted bar chart is usually clearer.
- Direct labels can reduce the effort required to match slices to a legend.
- Never present a percentage without clarifying its denominator, population, and time period when those details affect interpretation.

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(Some) Other Common Mistakes

% of what?

Missing units

Unclear time period

Using a legend with a pie chart (instead of direct labels)

Picking an inappropriate chart type

Be aware of your font size!



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ACTIVITY TIME!

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In small groups



Find an example of a terrible chart / data visualization

Identify 2+ reasons it's awful
What message does it convey?

Suggest 2+ improvements



Then, find an example of a wonderful, clear, insightful visualization:

Share 2+ reasons why you selected it

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WHAT ABOUT CHARTS MADE BY OTHER PEOPLE?

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When viewing visualizations:



Be critical.

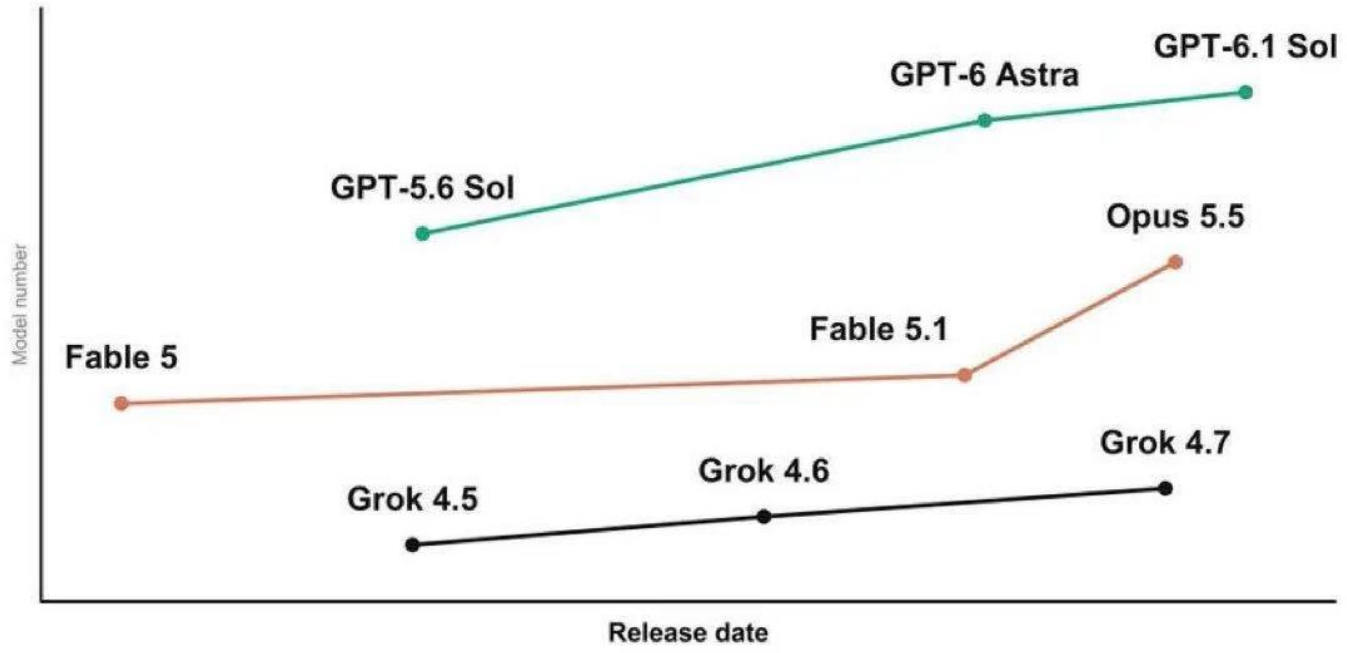


Be skeptical.

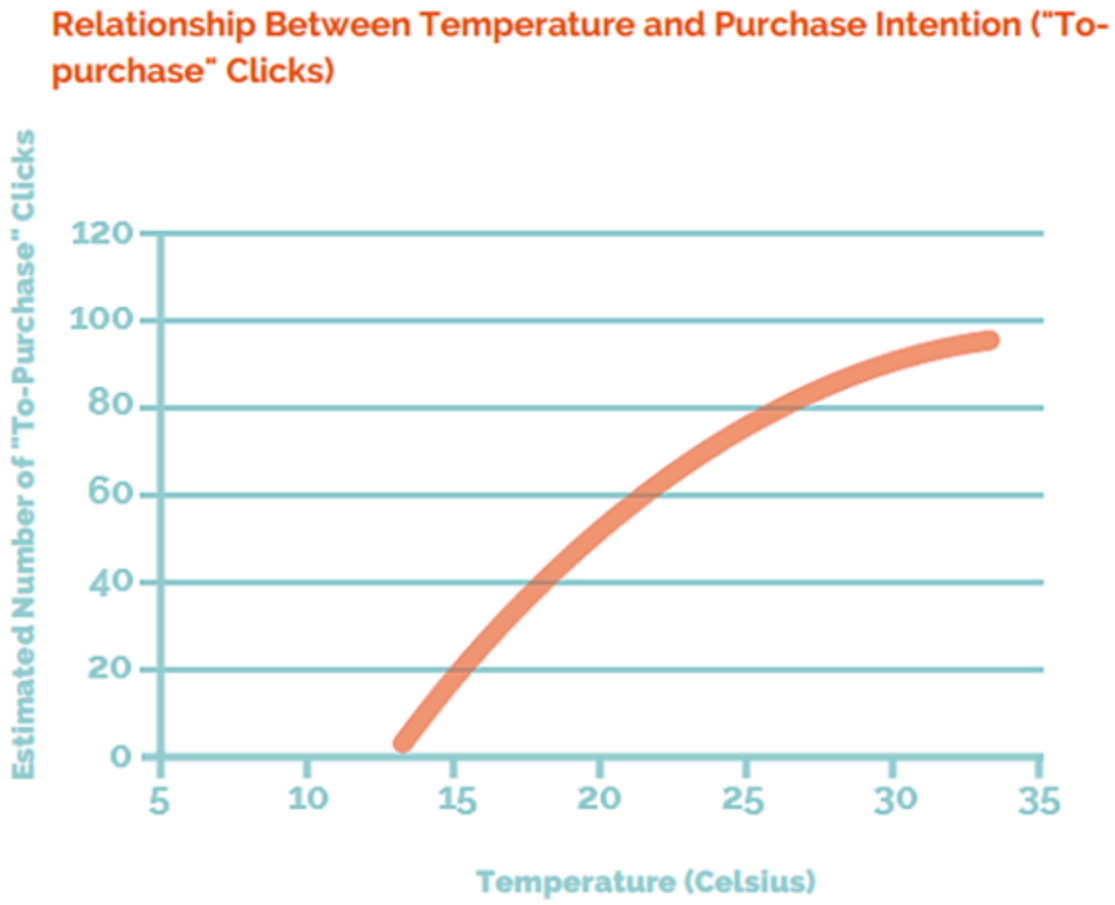
Pay attention to axes.

Sashi @sashikantsingh_ · 1d

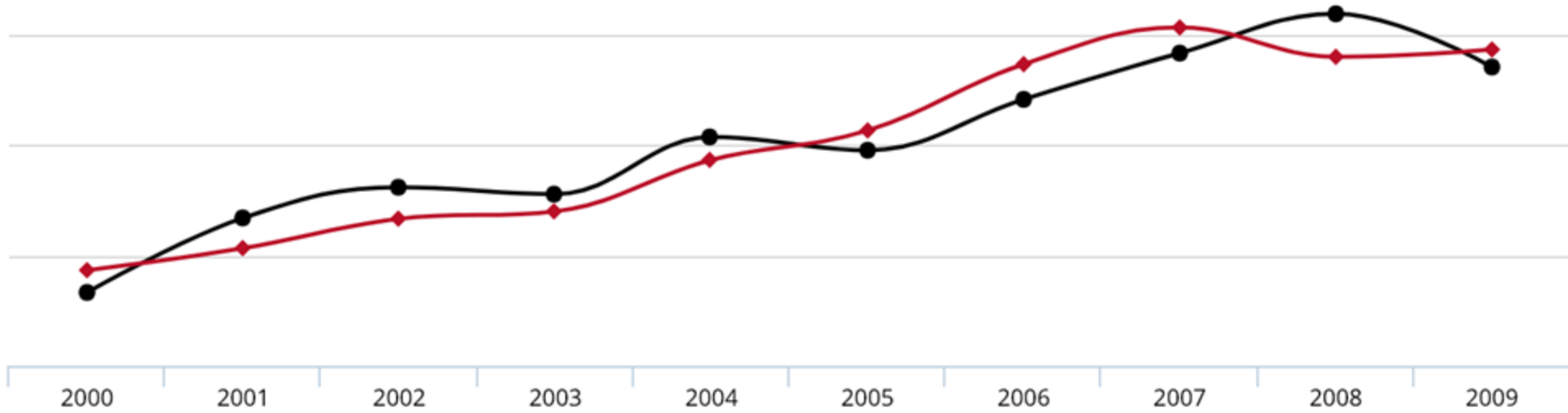
Open AI's internal benchmarks show GPT-6.1 Sol crushing Opus 5.5, with Anthropic struggling to keep up



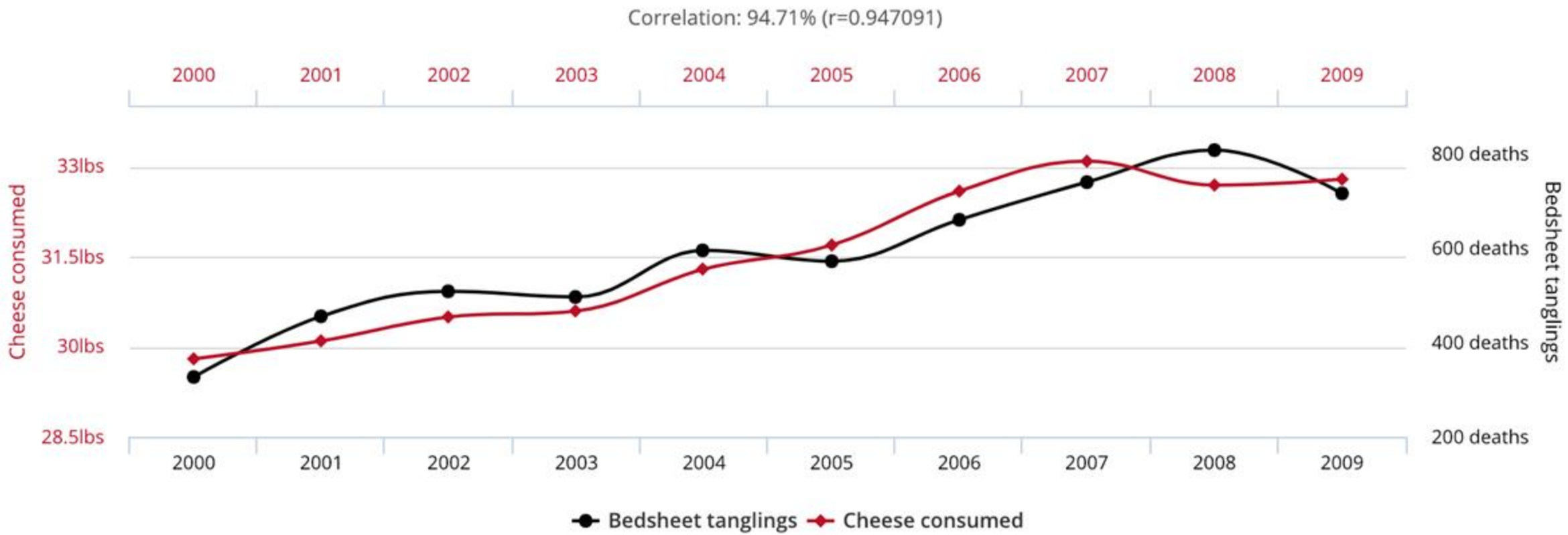
Be skeptical about correlation.



Source: 'The temperature premium: Warm temperatures increase product valuation' in Journal of Consumer Psychology (2013)


2000
2001
2002
2003
2004
2005
2006
2007
2008
2009

Per capita cheese consumption
correlates with
Number of people who died by becoming tangled in their bedsheets



Data sources: U.S. Department of Agriculture and Centers for Disease Control & Prevention

Misleading correlation

- Two variables can **move together** without one **causing** the other.
  - A strong correlation does not establish causation.
- A correlation may be **coincidental**, driven by a **third variable**, or influenced by the way the **data** was collected.
  - Outliers can strongly influence some correlation measures, particularly Pearson's correlation coefficient.
  - Examine scatter plots and the underlying observations rather than relying on a **single** correlation statistic.
- Consider whether the relationship is **plausible**, whether other variables could explain it, and whether the **evidence supports** a causal claim.

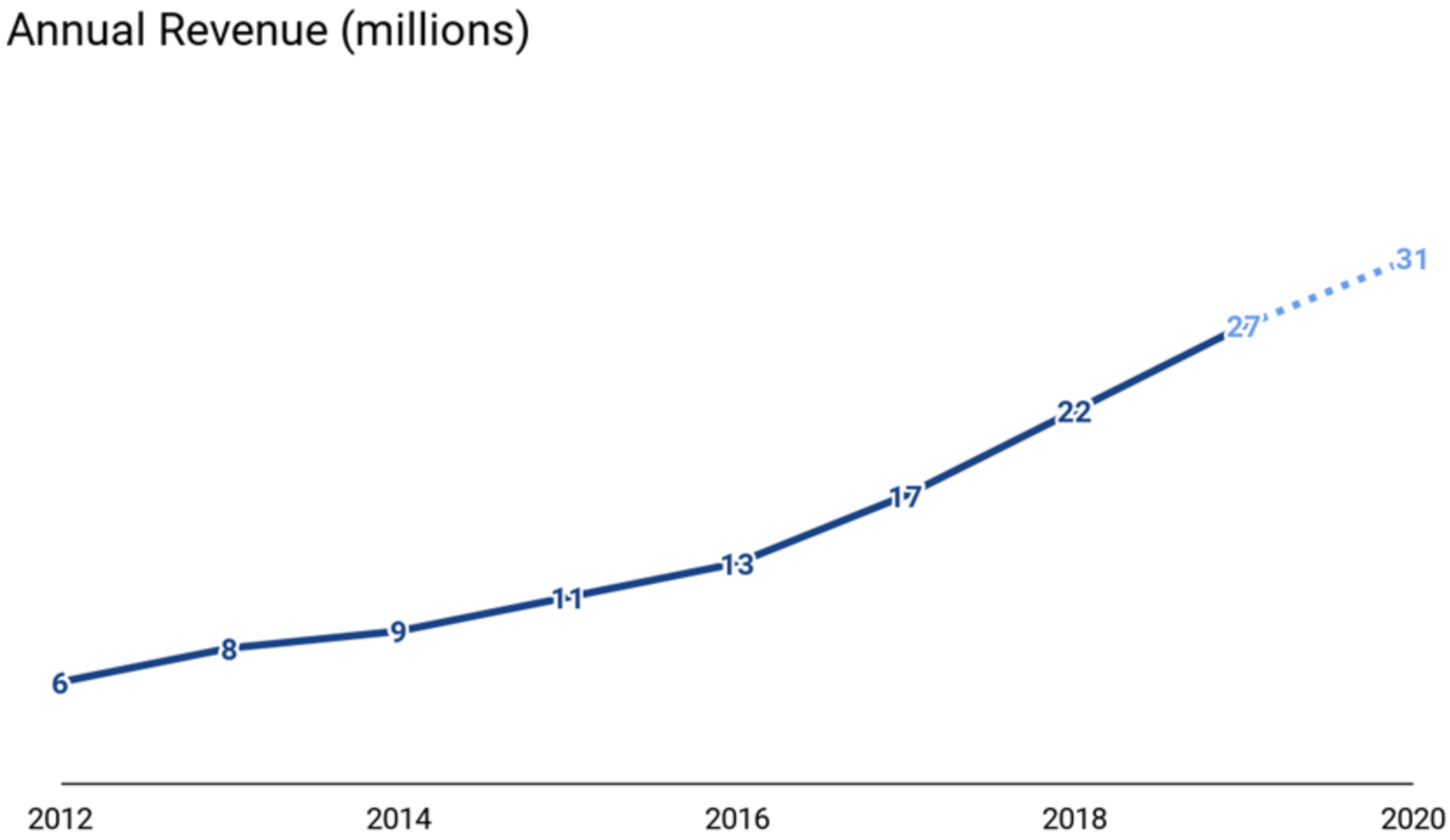
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Example: Ice cream and shark attacks

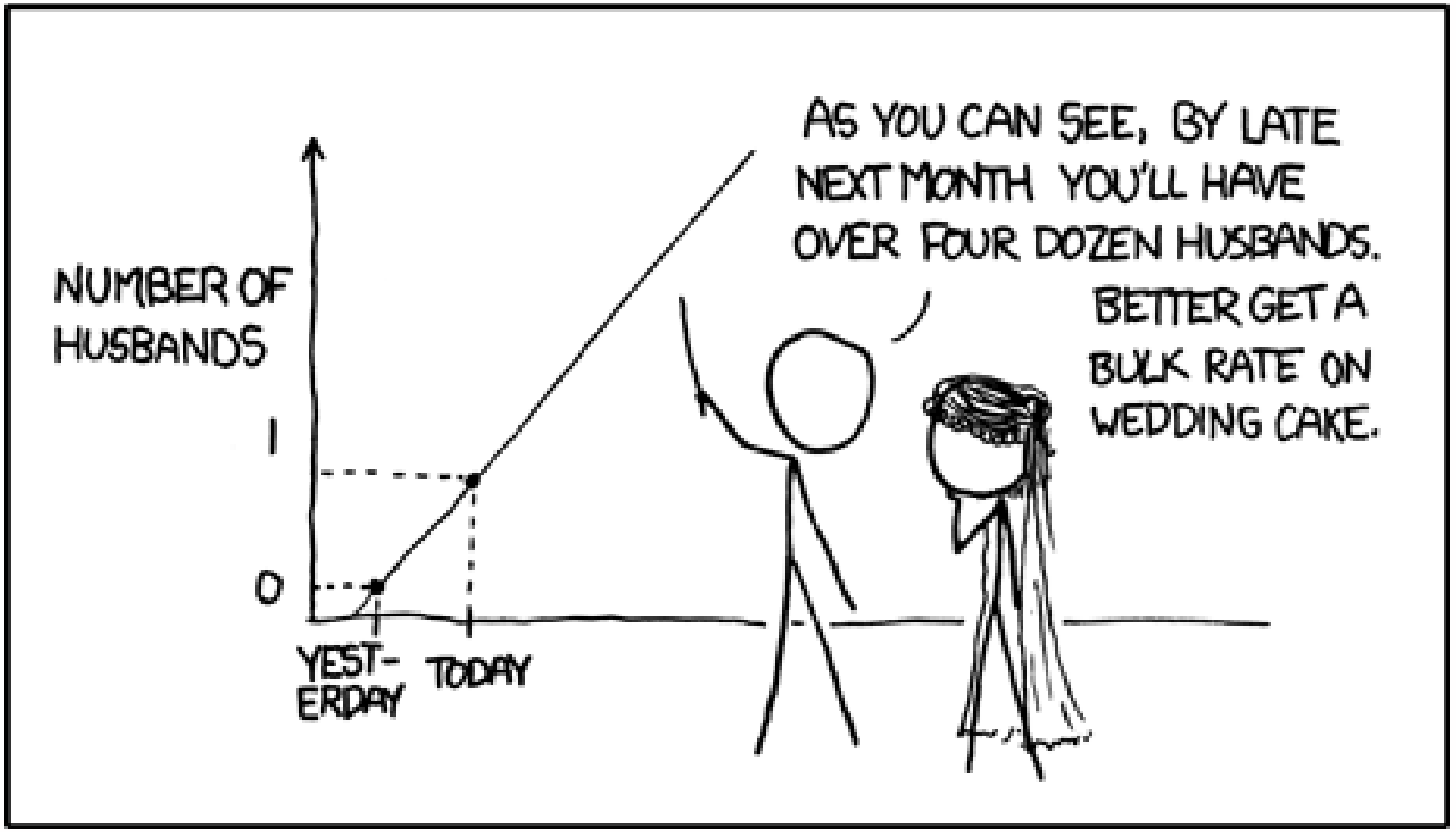
- Suppose **ice cream sales** and **shark attacks** both increase during summer.
- A misleading interpretation would be: "**Eating ice cream causes shark attacks.**"
- A more plausible explanation is that **warmer weather** increases both ice cream consumption and swimming activity.
- The visualization can accurately show a **relationship** while the accompanying narrative invents an unsupported explanation.

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Extrapolate reasonably



MY HOBBY: EXTRAPOLATING



Unreasonable extrapolation

- Extrapolation **extends an observed pattern** beyond the range of available data.
- A trend that held in the **past** may not continue into the **future**.
- Forecasts depend on **assumptions** about growth, seasonality, market conditions, and other factors.
  - Projections should identify their assumptions and distinguish observed values from forecast values.
- **Uncertainty generally increases** as a forecast extends further into the future.
- A trend line is a model of a **relationship**, not proof that the same relationship will continue indefinitely.

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Example: The hockey-stick forecast

A company reports revenue of $1 million, $1.2 million, and $1.4 million over three years.

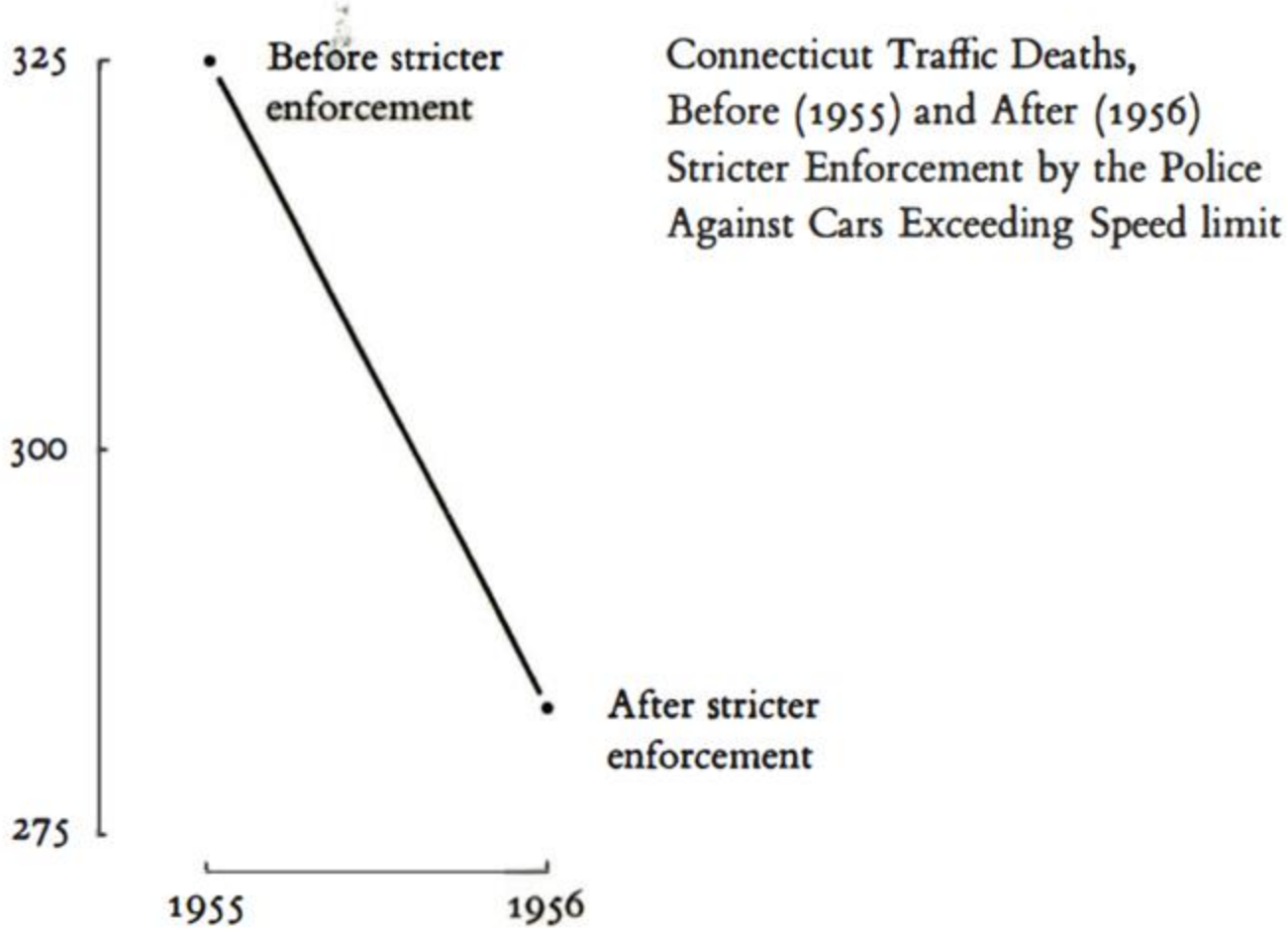
Its presentation projects revenue of $5 million three years later by extending the apparent upward trend.

- What **assumptions** support the projection?
- Is growth expected to remain **linear**?
- Is there **evidence** about market size, costs, competition, or capacity?
- Should the forecast show a **range** of possible outcomes instead of one precise line?

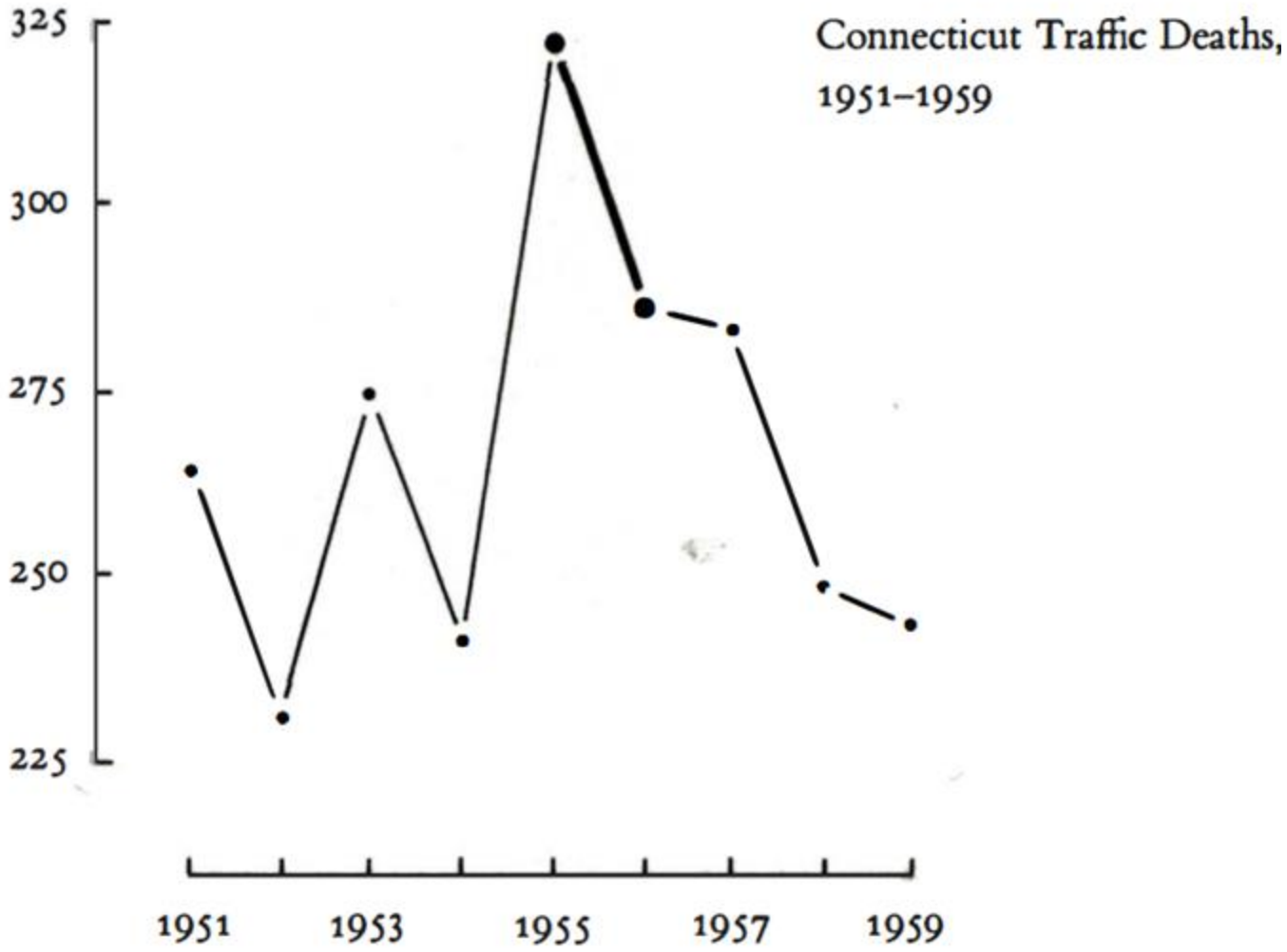
Forecasts can be **useful** even when **uncertain**. The problem is presenting an assumption-dependent projection as though it were an **observed fact**.

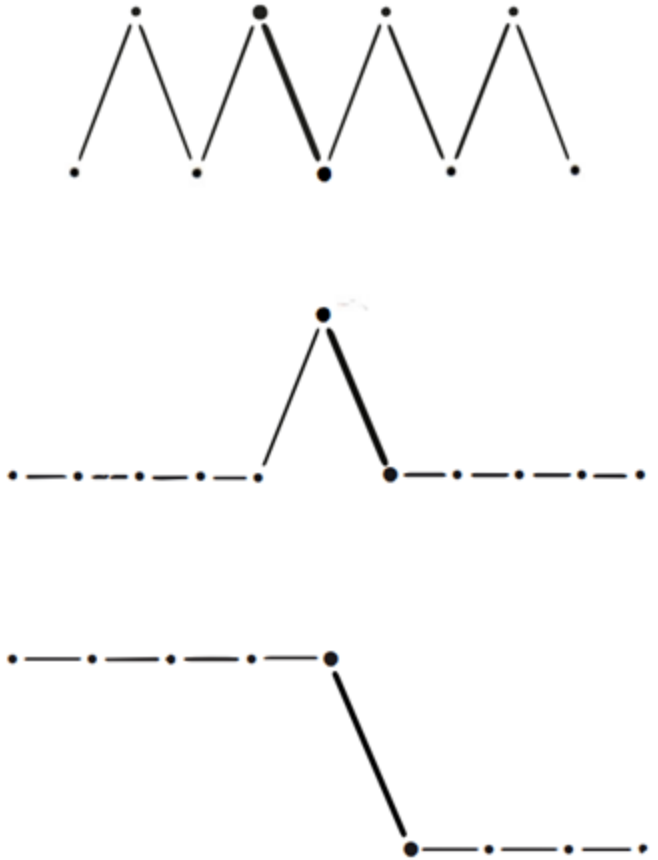
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Consider context.



Consider context.





Example: Traffic-data context

- A chart showing **traffic declining** over the last three months may suggest that a road redesign **reduced congestion**.
- But if traffic was **unusually high** during the starting month, or the chart omits **seasonal** patterns, the conclusion may be premature.
- Consider:
  - A longer historical baseline.
  - Comparable periods from previous years.
  - Changes in road closures, weather, or construction.
  - Whether traffic volume and travel time tell the same story.
- A carefully chosen time window can change the story without changing any individual observation.

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GIVING FEEDBACK

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Giving Constructive Feedback



Frame feedback around the artifact, not the person.



Expect subjectivity: it’s okay to see things differently!



Try to give at least 1 strength & 1 suggestion in each section.

The Sceptic Checklist

S	Scale: Are the axes, intervals, and baselines appropriate and clearly visible?
C	Context: Are the units, dates, population, definitions, and relevant comparisons provided?
E	Evidence: Is the conclusion supported by the data, or does it go beyond what is shown?
P	Proportions: Are counts, rates, percentages, and denominators being compared fairly?
T	Time: Is the time period appropriate, complete, and comparable?
I	Interpretation: Does the visualization distinguish association from causation and observation from prediction?
C	Clarity: Are the chart type, labels, colour choices, and annotations helping the audience understand the data?

Activity: Peer Review



Review your partner's visualization for

- Design choices
- Data clarity
- Connection to business question



Share & discuss feeback



Debrief as a class: themes, successes, challenges.

Peer Review Focus Areas

Can you tell what question this viz is trying to answer?

Is the chart type appropriate for that question?

Is the design simple & clear? Anything confusing or misleading?

Are we avoiding too many chart types in one place?

Are axes and legends clearly labelled?

Can you quickly see patterns, comparisons, or trends?

Are colours meaningful and colour-blind friendly?

1-Minute Reflection



WHAT WORKED?



WHAT WILL YOU TRY DIFFERENTLY NEXT TIME?

Closing

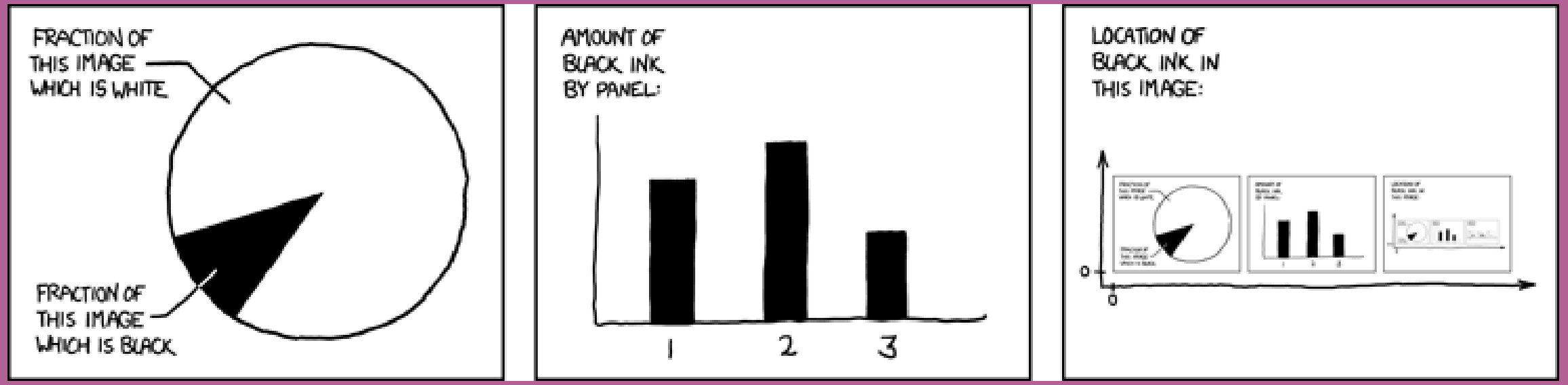
- 1-minute reflection: Why are visual lies often more convincing than numerical ones?

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What’s next?

- Bias in data analysis

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FRACTION OF THIS IMAGE WHICH IS WHITE
FRACTION OF THIS IMAGE WHICH IS BLACK
AMOUNT OF BLACK INK BY PANEL:
1
2
3
LOCATION OF BLACK INK IN THIS IMAGE:
0
0